

In vitro techniques to examine mTOR pathway in differentiated human astrocytes



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Abstract

Introduction: Since the advent of stem cell reprogramming technology, previously inaccessible live human astrocytes can now be generated for research, using a range of astrocyte differentiation protocols that have emerged over the past two decades. Mounting evidence indicates an active involvement of astrocytes in the pathophysiology of various neurological diseases, contrasting its previously assumed auxiliary role. As a key regulator of many growth-related cellular processes, mTOR pathway dysregulation results in a range of diseases, including tuberous sclerosis complex, which is a neurological disorder with evidence for astrocyte dysfunction arising from hyperactivation of the mTOR pathway. In this project, differentiation methods are trialled to produce human astrocytes, and assays to investigate astrocyte function as well as the effects of nutrient supply on mTOR pathway activation are optimised.

Methods: Two differentiation protocols (1. Using STEMCELL Technologies Kits and 2. Serum-driven differentiation) were evaluated for their suitability in generating human astrocytes. For optimisation of assays to study effects of nutrient supply on mTOR pathway activation, rat astrocytes were incubated in glucose- and serum-free media for 24 and 72 hours to investigate the change in pS6:S6 ratios. To optimise glutamate uptake assay, glutamate uptake was compared between neuron-astrocyte co-cultures and astrocyte mono-cultures to assess the effect of neuronal co-culturing on glial function.

Results: Neither astrocyte differentiation protocols generated mature human iPSC-derived astrocytes. Starvation of both glucose and serum did not lead to a significant decrease in mTOR pathway activation; however, serum starvation significantly impaired S6 expression in rat astrocytes for both incubation periods (24 hours: $P=0.0262$, $n=3$; 72 hours: $P<0.0001$, $n=3$). Co-cultures showed a higher rate of glutamate uptake than astrocyte mono-cultures ($P=0.0497$, $n=3$).

Conclusion: To establish a suitable method to generate human mature astrocytes, further rounds of astrocyte differentiation should be conducted. Other differentiation protocols could also be explored for standardisation of differentiation strategies. More endeavours in researching the mTOR pathway and astrocytes will hopefully increase the understanding of their roles and uncover potential therapeutic targets for a range of diseases such as TSC.

Abbreviations

Alzheimer's disease (AD), astrocyte differentiation medium (ADM), astrocyte induction medium (AIM), astrocyte maturation medium (AMM), astrocyte maintenance medium (astro main), central nervous system (CNS), coronavirus disease 2019 (COVID-19), coefficient of variation (CV), 4',6-Diamidino-2-Phenylindole (DAPI), dibutyryl cyclic AMP (dBcAMP), dual SMAD inhibition (dSMADi), excitatory amino acid transporter (EAAT), embryoid body (EB), embryonic stem cells (ESCs), fluorescence-activated cell sorting (FACS), foetal bovine serum (FBS), glial fibrillary acidic protein (GFAP), Hank's Buffered Salt Solution (HBSS), human leukaemia inhibitory factor (hLIF), human embryonic stem cells (hESCs), hypoxanthine phosphoribosyltransferase (HPRT), immunofluorescence (IF), induced pluripotent stem cells (iPSCs), knock-out (KO), medium change (MC), mechanistic target of rapamycin (mTOR), mTOR complex 1 (mTORC1), neurogenin2 (NGN2), neurogenin2 medium 1 (NGN1 medium), neurogenin2 medium 2 (NGN2 medium), neural induction medium (NIM), neural maintenance medium (NMM), neural progenitor cells (NPCs), neural stem cells (NSCs), phosphate-buffered saline (PBS), paraformaldehyde (PFA), protease and phosphatase inhibitor (PPI), pluripotent stem cells (PSCs), phospho-S6 (pS6), p70 ribosomal S6 kinase (p70S6K), penicillin-streptomycin (P/S), quantitative real-time PCR (qRT-PCR), radial glial cells (RGCs), p90 ribosomal S6 kinase (RSK), SB431542 (SB), S6 ribosomal protein, subependymal giant cell astrocytomas (SEGAs), tuberous sclerosis complex (TSC), wild-type(WT)

1 Introduction

1.1 Stem cell research

The advent of induced pluripotent stem cells (iPSCs) from ongoing stem cell research has received increasing attention from the many unprecedented opportunities they provide for basic cellular science, pharmacology, and therapeutics.

In 2006, Shinya Yamanaka successfully produced iPSCs from mouse fibroblasts by introducing four transcription factors, Oct-3/4, Sox2, KLF4, and c-Myc (Takahashi & Yamanaka, 2006). Since, further research has examined comparability with embryonic stem cells (ESCs), which are the natural pluripotent stem cells (PSCs), improved stem cell culturing methods by quality control and teratoma assays, and directed differentiation to generate cultures enriched in cell types of a certain lineage by manipulating the extracellular microenvironmental conditions (Yamanaka, 2012; Zakrzewski et al., 2019).

The ability to reprogramme adult stem cells and somatic cells into iPSCs provides an alternative to obtaining PSCs from an embryo at the blastocyst stage. This, in turn, enables stem cell therapy and regenerative medicine without resorting to embryonic cells, which has been an ethically contentious issue for years. iPSCs have also revolutionised disease modelling by providing an *in vitro* human cell-based model for carrying out drug screening and toxicity tests (Rowe & Daley, 2019). This has been particularly meaningful for the study of neurological diseases where primary cultures are difficult to acquire, as human brain tissues can otherwise only be obtained from brain surgery or autopsies. In addition to the inaccessibility of live tissues to study neurological diseases, the significant differences in neural development between humans and other species results in limitations for animal models to accurately reflect neural conditions in humans. For example, cortical tuber development in individuals with tuberous sclerosis complex disease (TSC) has not been observed in rodent brains (Blair & Bateup, 2020). Furthermore, murine astrocytes have shown transcriptional, molecular and functional differences from human astrocytes (Zhang et al., 2016). Therefore, stem cell reprogramming technology has provided new possibilities in multiple branches of scientific research with great expectations as a promising tool for developing more accurate *in vitro* models.

1.2 Astrocytes

Making up 20-40% of all glial cells, astrocytes are one of the largest glial cell populations in the central nervous system (CNS) and play important roles in brain physiology and function (Ji et al., 2019).

Understanding of astrocytic functions and roles

Historically, astrocytes were thought to be auxiliary "inferior" cells providing mechanical and trophic support for neurons. They were later appreciated to also be involved in a range of essential brain functions. These include maintaining the structural integrity of the blood-brain barrier, maintaining chemical homeostasis by regulating neurotransmitter recycling, signal transmission within and between neuronal-glial networks, as well as the formation and function of neuronal synapses (Ji et al., 2019). One of the more important and established astrocytic functions is glutamate uptake, as it protects neurons from glutamate excitotoxicity by removing glutamate from the synapse to maintain a constant low level of extracellular glutamate concentrations (Brown, 1999; Danbolt, 2001).

A manifestation of the shifting view of the nervous system is the proposal of the tripartite synapse model which accounts for the active role of astrocytes in regulating synaptic transmission. Tripartite synapse theory entails that synaptically-associated astrocytes are not only in close physical proximity with the neuronal synapse, but show also intimate functional involvement (Ji et al., 2019).

Heterogeneity of astrocytes

Astrocytes were initially described as a homogeneous cell population, but accumulating evidences show a high morphological and functional heterogeneity of astrocytes within and across brain regions and throughout the CNS (Krencik & Zhang, 2011; Tabata, 2015). Specific markers preferentially stain astrocytes in certain brain regions, potentially corresponding to differences in morphological and functional characteristics. For example, glial fibrillary acidic protein (GFAP) has been reported multiple times to be more suitably used for staining hippocampal than striatal astrocytes (Matias et al., 2019); whereas S100 β is more suited for labelling astrocytes in the cortex and thalamus (Zhang et al., 2019). Zhang et al. (2019) have also claimed that there is currently not one marker that labels all astrocyte subpopulations successfully.

Astrocyte dysfunction

It is increasingly appreciated that astrocytes play an active role in the pathophysiology of neurological diseases (Ji et al., 2019; Julia et al., 2017; Matias et al., 2019). For example, in response to an insult or injury, astrogliosis occurs wherein astrocytes switch into the reactive state and migrate to the nearby site of neuronal destruction to enable repair as part of the inflammatory cascade. However, prolonged reactive astrogliosis is problematic and is a common phenomenon found in a number of CNS disorders, such as Alzheimer's disease (AD), epilepsy, and Parkinson's disease (Liu et al., 2017).

Other aspects of astrocyte dysfunction found in CNS disorders include intracellular accumulation of A β in astrocytes as well as downregulation of EAAT2 affecting glutamate clearance in AD, altered glutamate homeostasis and potassium buffering inducing excitotoxicity in epilepsy (Liu et al., 2017).

1.3 Astrocyte differentiation

One of the significant benefits of the emergence of stem cell reprogramming technology is the ability to generate previously unattainable human cell types for studies. This enables neurological disease models to investigate human astrocyte dysfunction, and thus providing insight into the mechanisms, biology and development of astrocytes in disease. Moreover, establishing these human ESC (hESC)-derived disease models and their isogenic controls by genetically modifying hESCs containing the mutated genes with CRISPR/Cas9 technology allows the testing of hypothesis on the causal role of specific neurological disease mutations. Neurological disease models also allow preclinical testing of potential therapeutics and their toxicity, as well as identify critical windows during development for treatment (Saber & Sahin, 2020).

Astrocyte differentiation protocols

Differentiation of astrocytes generally starts from PSCs developing into neural stem cells (NSCs), also known as radial glial cells (RGCs), then differentiating into glial progenitor cells (GPCs), and finally into mature astrocytes. *In vitro* this can be achieved by manipulation of cell culture conditions by supplementation of a series of different growth factors at each developmental stage. Progression to a particular cell fate is determined by expression of different markers depending on the stage of maturity (Akdemir et al., 2020; Engel et al., 2016).

Chandrasekaran et al. (2016) have summarised astrocyte differentiation protocols developed so far over the past two decades. These protocols are categorised into two major classes regarding the way in which NSCs are formed: an embryoid body (EB)-based technique and a monolayer-based dual SMAD inhibition (dSMADi) method. A schematic of differentiation of astrocytes with the EB-based technique is shown in Figure 1. The monolayer-based dSMADi technique described in Shi et al. (2012) is used in our laboratory to differentiate PSCs into NSCs by neural induction via dSMADi, which is also the method used for NSC differentiation in this project.

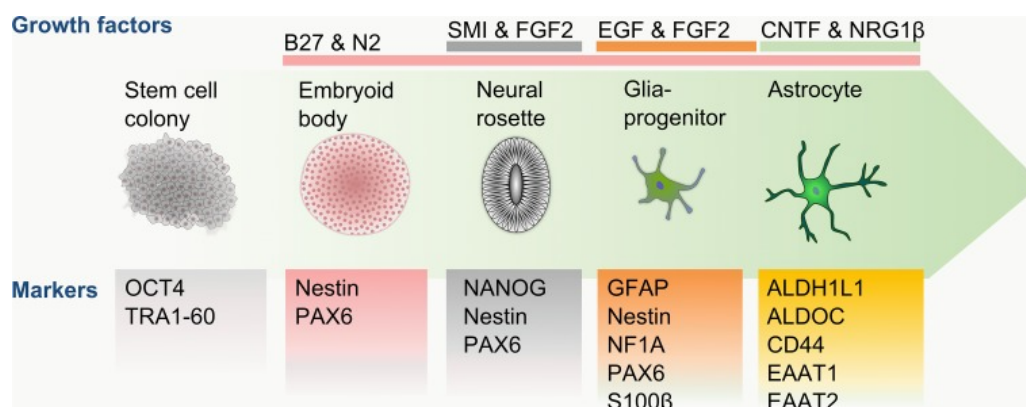


Figure 1: Schematic diagram of embryoid body-based astrocyte differentiation showing growth factors used and markers expressed at different stages (Engel et al., 2016)

Astroglial markers

Different genes show varying expression at different developmental stages, thus they are crucial markers to indicate the maturation level of cultured cells during differentiation *in vitro*. Although there have not been a standardised set of criteria for defining astrocytes and their subtypes, there are a range of astroglial markers used across the literature, summarised in Chandrasekaran et al. (2016). Four established markers (EAAT1, EAAT2, GFAP, and S100β) are chosen to assess the astrocytes generated from differentiation protocols (Table 1).

Probably the most well-established astrocytic marker, GFAP, is an intermediate filament protein highly expressed in mature astrocytes, providing mechanical support to astrocytes as part of the cytoskeleton (Szymaś, Tokarz, et al., 1986). However, it is also expressed by RGCs in the NSC stage, which is distinct from the astrocyte stage. Thus, GFAP expression could indicate that cells in differentiation are in the developmental stage of GPCs or mature astrocytes (Mamber et al., 2012). S100β calciprotein expression defines a later developmental stage, as it was shown to be expressed in only GFAP-expressing cells that had matured and lost their NSC potential (Raponi et al., 2007). EAAT1 and EAAT2 are excitatory amino acid transporters (EAATs) principally expressed in astrocytes at their early and late stages respectively. As sodium-dependent glutamate/aspartate symporters, they are responsible for mediating synaptic glutamate clearance in astrocytes (Danbolt, 2001). EAATs maintain a low extracellular glutamate level and prevent neurotoxicity resulting from hyperstimulation of neurons due to excess glutamate (Rothstein et al., 1996).

Marker	General stage of presentation
EAAT1	Early astrocytes
EAAT2	Late postnatal astrocytes
GFAP	Early GPCs, late astrocytes
S100 β	Late GPCs

Table 1: Astroglial markers and their stage of presentation

Challenges and limitations

Astrocyte differentiation has at least three limitations: the lack of standardisation in differentiation strategy, the lack of standardisation of defined astrocytic markers, and the heterogeneity of astrocytes generated.

There is a wide variability in astrocyte differentiation protocols with regard to the reported efficiency of astrocyte generation, the growth factors applied, and the accepted percentage purity of an astrocytic culture (Engel et al., 2016). The characterisation of astrocytes by their expression profile and functionality is also variable with a range of early developmental, late maturation markers, and physiological functions used to identify derived cultures across studies, leading to variable differentiation outcomes (Chandrasekaran et al., 2016; Engel et al., 2016).

Reflective of the relative novelty of the field, this lack of standardisation and the little knowledge in astrocyte subtypes are challenges in using differentiation strategy to derive human astrocytes from PSCs. Astrocyte heterogeneity further complicates the analysis, interpretation and comparison of differentiation outcomes.

Given the continually growing interest in astrocytes across the neuroscience field with the discovery of its central role in modulating healthy and abnormal brain development and function (Chandrasekaran et al., 2016), it is hopeful that standardised protocols will be developed for *in vitro* astrocyte generation with more clarity in maturity status and phenotypic properties in the near future.

1.4 mTOR pathway

As the master regulator of cell proliferation and survival, metabolism, and ageing, the mechanistic target of rapamycin (mTOR) pathway is an important signalling cascade integrating external environmental cues and signal inputs from multiple other pathways, such as Akt and ERK pathways, to cause intracellular changes in a range of main growth-related cellular processes. They include translation initiation, ribosome biogenesis, lipogenesis and glycolysis (Aramburu et al., 2014; Dazert & Hall, 2011) .

When the mTOR pathway is activated, mTOR complex 1 (mTORC1), which is a key signalling molecule in the pathway, stimulates a number of downstream proteins. One of the activated proteins is p70 ribosomal S6 kinase (p70S6K), which in turn phosphorylates and converts the constitutively expressed S6 ribosomal protein (S6) into phospho-S6 (pS6), another key positive regulator of mTOR pathway and translational processes. Hence, the overall effect of mTORC1 activation promotes protein synthesis and cell proliferation, inhibits autophagy and apoptosis (Feng et al., 2005; Portocarrero et al., 2018). A schematic of the mTORC1 signalling pathway is shown in Figure 2. In addition, mTORC1 activation promotes phosphorylation of S6; therefore, pS6:S6 ratio is useful as a measure of degree of mTOR pathway activation.

With its involvement in a multitude of cellular functions, it is unsurprising that mTOR dysregulation is implicated in a wide variety of diseases including cancers, metabolic disorders, and neurological diseases, such as TSC (Dazert & Hall, 2011).

As the mTOR pathway becomes activated to a different degree depending on surrounding signals, processes regulated by mTOR pathway are subject to changes in the cellular environment. For example, in response to stress, such as hypoxia and the lack of nutrients, cells normally limit proliferative and growth-related processes by altering signal transduction and gene expression to adjust to the modified environment to have a higher chance of survival (Aramburu et al., 2014).

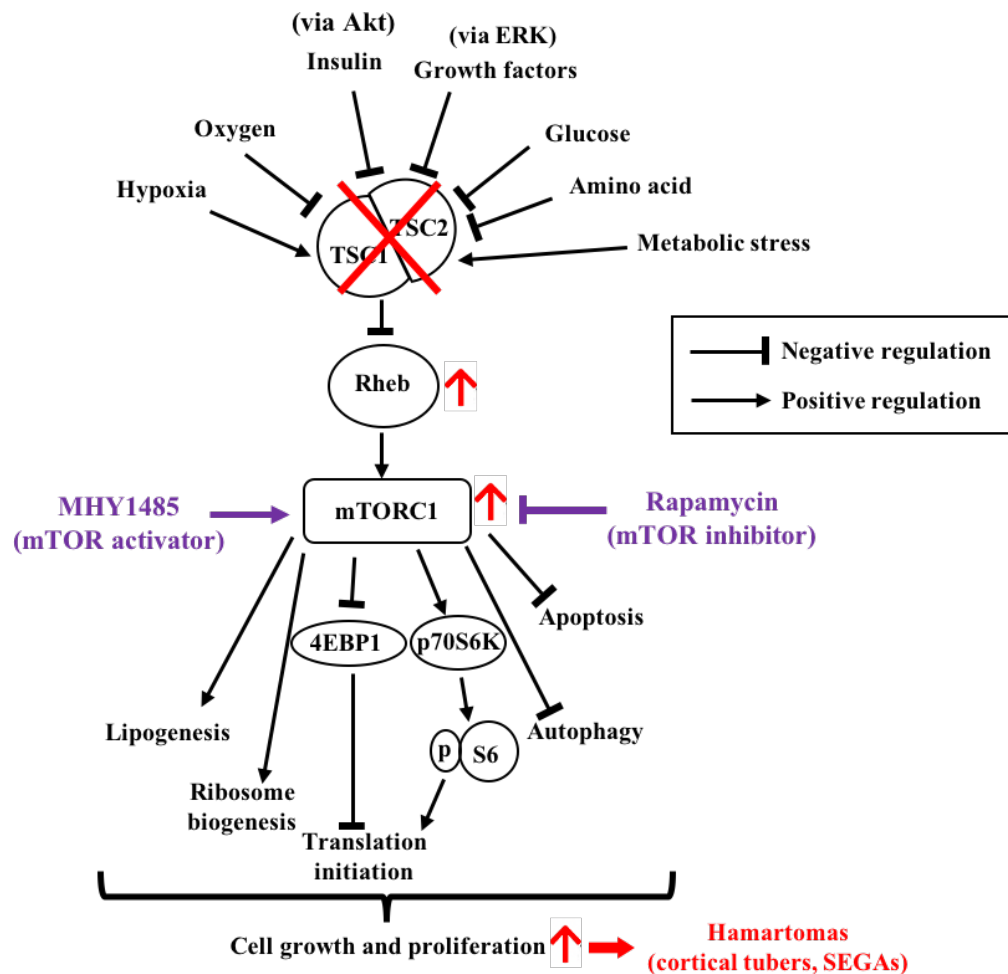


Figure 2: Schematic diagram of mTOR signalling pathway and proposed alteration in TSC
 A schematic of the mTORC1 signalling pathway (black) and how it is altered by a dysfunctional TSC1:TSC2 complex in the proposed disease mechanism of TSC (red). The mTOR pathway can be activated by an mTOR activator (MHY1485), or inhibited by an mTOR inhibitor (rapamycin), indicated in purple arrows.

mTORC1 activation in response to nutrient supply

As a mechanism of nutrient intake to promote cell growth and proliferation, metabolic signals and supply of nutrients like glucose and amino acids increase mTORC1 activation indicated by the increased S6 phosphorylation (Feng et al., 2005; Haselbacher et al., 1979; Thomas et al., 1982). Conversely, studies have indicated that serum activation of p90 ribosomal S6 kinase (RSK) could lead to S6 phosphorylation via Ras/ERK signaling cascade independent of the mTOR pathway, leading to similar cellular consequences (Roux et al., 2007). Therefore, modelling the mTOR pathway with an mTOR activator (MHY1485) or mTOR inhibitor (rapamycin) is crucial to confirm whether the observed cellular phenotypes are dependent on the mTOR modification.

1.5 Tuberous sclerosis complex

Being one of diseases with indication of a dysregulated mTOR pathway, TSC is a rare neurodevelopmental disorder caused by inactivating mutations in at least one of the two autosomal tumour suppressor genes (TSC1/2). It is characterised by the formation of benign tumours called hamartomas in multiple organs commonly including the brain (Wong & Crino, 2012), resulting in various neurological manifestations, the most common of all being epilepsy (60 - 90%) (Orlova & Crino, 2010).

TSC is a condition with a significant unmet clinical need (Blair & Bateup, 2020). This is because conventional treatments with rapalogues (rapamycin and its derivative everolimus), which are mTOR inhibitors, despite showing effectiveness, remain highly toxic for chronic use (Portocarrero et al., 2018).

Proposed disease mechanism of TSC

Upstream of mTORC1, the TSC1:TSC2 complex negatively regulates the mTOR pathway. A dysfunctional TSC1:TSC2 complex results in a loss of inhibition on Rheb, which normally activates mTORC1. Disinhibited Rheb thus hyperactivates mTORC1 and lead to downstream consequences including increased cell proliferation and the formation of hamartomas (Feng et al., 2005) (Fig 2).

Haploinsufficiency of TSC1 or TSC2 is sufficient to result in a dysfunctional TSC1:TSC2 heteromeric complex, but the homozygous knockout genotype is associated with a more severe phenotype than the heterozygous genotype; additionally, TSC2 mutations exhibited more severe phenotypes compared to TSC1 mutations in humans and mouse TSC models (Magri et al., 2013).

Astrocytes in TSC

Brain hamartomas in TSC, such as cortical tubers and subependymal giant cell astrocytomas (SEGAs) characteristically consist of increased numbers of enlarged abnormal astrocytes, resulting in glioneuronal imbalance (Wong & Crino, 2012). Apart from the biological standpoint, tubers also invariably display severe astrogliosis, comprising of a mixture of gliotic and reactive astrocytes that are functionally compromised with impaired potassium buffering and glutamate clearance (Sosunov et al., 2008), which likely promote neuronal excitability and seizures in TSC. One central question for investigation is whether the development of astrogliosis in tubers, in favour of epileptogenesis, is directly linked to TSC1/2 mutations in astrocytes (Sosunov et al., 2008; Wong & Crino, 2012).

1.6 Project aims

The project originally set out to test the hypothesis that astrocytes have a central role in TSC, as previous studies have identified correlations between astrocyte dysfunction, formation of hamartomas, and epileptogenesis in TSC. Nonetheless, an *in vitro* system and assays need to be first developed and optimised in order to examine the mTOR pathway as well as the glial functioning in human astrocytes in the context of TSC. Overall, three aims are identified in this project:

1) Compare the suitability of two astrocyte differentiation protocols for generation of human astrocytes

The following astrocyte differentiation protocols are chosen to be trialled for their suitability in generating mature human astrocytes:

1. Astrocyte differentiation using STEMCELL Technologies Kits
2. Serum-driven differentiation method

The STEMCELL Technologies Kits are commercially available, but the formula of basal media and supplements are unknown. As for the serum-driven differentiation protocol, the strategy is principally based on the addition of leukaemia inhibitory factor (LIF) and serum to the neuronal medium for differentiation and maturation of inflammation-responsive astrocytes from the NSC stage (Santos et al., 2017). As one of the few studies reporting the generation of pro-inflammatory astrocytes, it is useful for modelling neurological diseases with an inflammatory component given the increasing indication of astrocytic role in neuroinflammation (Santos et al., 2017).

Hence, to establish a method to generate human astrocytes, the suitability of these differentiation protocols is first assessed using iPSCs, before selection of a more suited one for subsequent differentiation using TSC2 wild-type (WT) and KO hESCs.

2) Examine the effect of nutrient supply on mTOR pathway activation in astrocytes for assay optimisation

In the context of TSC, published work in the laboratory showed that high glucose concentrations mask cellular phenotypes in an iPSC model (Rocktäschel et al., 2019), in which TSC2 KO NSCs derived from human PSCs cultured in normal culture medium interestingly showed no phenotypic difference from TSC2 WT NSCs. The glucose level in standard NSC media is much higher than physiological glucose levels. It is possible that high glucose levels "maintained" the same outcome of

mTOR pathway in NSCs; hence, TSC2 KO cells only displayed their corresponding phenotypes when they are starved of glucose. Furthermore, this points to the notion that there is metabolic rerouting and likely glycolytic shifting in TSC-deficient cells due to mTORC1 hyperactivation. This leads to a vulnerability to glucose "addiction" (Medvetz et al., 2015). This motivates the investigation of whether such a glucose-interfering pattern also appears in TSC2 WT and KO astrocytes.

Therefore, in this project, the effect of nutrient supply (glucose and serum) on mTORC1 activation are first studied using rat astrocytes for assay optimisation.

3) Optimise glutamate uptake assay for studying astrocyte function

The glutamate uptake assay is an assay developed to measure the glutamate uptake in a culture using the enzymatic colourimetric glutamate assay for measuring its concentration in a tested sample. It could be very useful in studying astrocytes in TSC given the links between altered glutamate homeostasis, astrocyte dysfunction and pathophysiology of TSC. For example by examining the glutamate uptake function of hESC-derived TSC2 KO astrocytes, one can investigate whether TSC2 KO astrocytes show impaired glutamate uptake function, compared to WT astrocytes, and furthermore, whether it is correlated to decreased EAAT1/2 expression. This is because gliotic astrocytes found in human tuber tissues showed lower EAAT1/2 expression, which is conducive to a hyperexcitable extracellular environment associated with vulnerability to epileptic seizures (Talós et al., 2008).

The enhancing effect of glial co-culture to neuronal function and survival is widely accepted (Brown, 1999). On the other hand, it is unclear whether astrocytes also, in turn, benefit from neuronal presence. Previous studies have shown that neurons are responsible for regulating glutamate transporter expression in astrocytes (Swanson et al., 1997). The presence of neurons were also found to be important due to the release of vasoactive intestinal peptide for astrocytes to manifest its protective role to avoid glutamate toxicity (Brown, 2000). It could thus be hypothesised that the functional interaction between neurons and astrocytes at the synapse is not only bidirectional as a tripartite synapse, but their functions are also mutually dependent.

Therefore, the glutamate uptake assay is carried out on neuron-astrocyte co-cultures and astrocyte mono-cultures to study the effects of neuronal co-culturing on glial functioning in terms of astrocytic glutamate uptake for optimising the assay.

2 Materials and Methods

This section introduces the methods and techniques that were used to conduct the experiments. Protocols for differentiation and culturing different cell types are described in section 2.1. Section 2.2 - 2.5 explains the protocols for immunofluorescence (IF), qualitative real-time PCR (qRT-PCR), western blotting, and glutamate uptake assay, respectively. Section 2.6 describes the statistical tests used to analyse the data. Finally, section 2.7 includes the criteria used for comparing between astrocyte differentiation protocols trialled for their suitability in generating human astrocytes.

2.1 Cell culture

2.1.1 iPSC culture

Control SBAD-03 induced pluripotent stem cells (iPSCs) generated from the lab were thawed by warming a cryovial in a 37 °C water bath for maximum 2 minutes, followed by centrifugation (400 g, 5 minutes) in 9 ml phosphate-buffered saline (PBS) (Gibco, 14190-094). The cell pellet was resuspended with mTeSR (STEMCELL Technologies, 85850) supplemented with 10 µM Rho kinase inhibitor Y-27632 (Abcam, ab120129) and were plated onto well surfaces coated with matrigel at 80000 cells/cm². Matrigel coating was made up by diluting one aliquot of BD Matrigel hESC-qualified Matrix (BD Biosciences, 354277) in 25 ml of cold knockout-DMEM (Thermo Fisher, 10829018) according to the manufacturer's instructions. Prior to plating, 100 µl/cm² matrigel was added to the wells and incubated at 37 °C for 1 - 20 hours before use.

iPSC cultures were continuously maintained with daily full medium changes (MCs) of mTeSR without Rho kinase inhibitor. At full confluency, iPSCs were passaged with the regular EDTA method: first, incubating cells in 0.1 ml/cm² EDTA (Invitrogen, 15575020) at 37 °C for 4 minutes after removal of culture medium, then EDTA was aspirated and cells were lifted by rinsing with the culture medium followed by re-plating of iPSCs on matrigel-coated wells.

2.1.2 Rat astrocyte culture

Rat Primary Cortical Astrocytes (Invitrogen, N7745100) were thawed and plated according to the manufacturer's protocol (Thawing and Establishing Rat Primary Cortical Astrocytes, MAN0001679) onto non-coated wells at 20000 cells/cm² using normal rat astrocyte medium (Table 2). Passaging and expansion upon full confluency was also carried out according to the manufacturer's protocol (Expanding Rat Primary Cortical Astrocytes, MAN0001680).

In the study of effect of nutrient supply on mTOR pathway activation, rat primary astrocytes were cultured in glucose-free, and serum-free culture media. See Table 2 for details of medium composition.

Condition of rat astrocyte media	Glucose	Serum	Composition of media
Normal (control)	+	+	Dulbecco's Modified Eagle Medium (DMEM)/F12 GlutaMAX (Thermo Fisher, 31331028), 15% foetal bovine serum (FBS) (Life Technologies, 10500064), 100 U/ml P/S (Thermo Fisher, 15140148)
Glucose-free	-	+	DMEM with no glucose (Thermo Fisher, 11966025), 15% FBS, 100 U/ml P/S
Serum-free	+	-	DMEM/F12 GlutaMAX, 100 U/ml P/S

Table 2: Conditions of rat astrocyte media

2.1.3 Cell differentiation

2.1.3.1 NSC differentiation from iPSCs iPSCs were differentiated into NSCs by neural induction via dSMADi according to Shi et al., 2012.

Upon attaining a fully confluent monolayer of pluripotent cells, the culture medium was changed from mTeSR into neural induction medium (NIM) containing freshly added 10 μ M SB431542 (SB) (Tocris, 1614) and 1 μ M dorsomorphin (Tocris, 3093), responsible for dSMADi to allow ectoderm formation. Full daily MC with NIM was carried out starting from day 1 until neuroepithelial sheet cells appeared on around day 10-12.

Neuroepithelial sheet cells were passaged in clumps using dispase (Gibco, 17105-041): First, 0.2 mg/cm² dispase was added directly to culture medium and incubated at 37 °C for 7-9 minutes until the edges of the sheets started to lift up. Then, the sheets were gathered in a tube by flushing using fresh neural maintenance medium (NMM) with minimal pipetting to avoid breakage of the neuroepithelial sheets. After that, 10 μ l DNase from the RNase-Free DNase Set (Qiagen, 79254) was added followed by centrifugation at 300 RPM for 3 minutes. The pellet of cell clumps (collected from 2 wells of a six-well plate) was washed with 7 ml fresh NMM twice before repeating centrifugation (300 RPM, 3 minutes). The cell pellet was resuspended in NIM and plated onto laminin-coated wells. Surfaces were coated with 2 μ g/cm² laminin (Sigma, L2020) by adding 0.1 ml/cm² of 20 μ g/ml laminin solution diluted in PBS for 4 - 24 hours at 37 °C before plating. The next day, the culture medium was switched to NMM freshly supplemented with 20 ng/ml FGF-2 (Peprotech, AF-100-18B), exposing the cells to FGF-2 for 4 days before stopping FGF-2 treatment on around day 16. Passaged neuroepithelial sheet cells were continually maintained with NMM MCs every other day until the appearance of neural

rosettes on around day 18. Neural rosettes were dispase-passaged in NMM. The passaged neural rosettes were then subject to MCs with NMM every other day until around day 22. At this point, cells should have reached the NSC stage. Details of MCs for NSC differentiation and composition of NIM and NMM are listed in Table 3.

Day 0:	100% confluent monolayer of iPSCs
Day 1 - 10:	Full MCs with NIM daily
Day 11:	Passaged neuroepithelial sheet cells in clumps with dispase onto laminin-coated plates in NIM
Day 12:	Switched to NMM + FGF-2
Day 14:	Full MC with NMM + FGF-2
Day 16:	Full MC with NMM
Day 18:	Passaged neural rosettes in clumps with dispase onto laminin-coated plates in NMM
Day 19:	Full MC with NMM
Day 21:	Full MC with NMM
Day 22:	NSCs
NIM:	NMM + SB + dorsomorphin
NMM:	1:1 mixture of DMEM/F12 GlutaMAX:Neurobasal medium (Thermo Fisher, 12348017), 1% B27 (Life Technologies, 17504044), 0.5% N2 (Life Technologies, 17502048), 1 mM L-glutamine (Life Technologies, 25030024), 500 μ M sodium pyruvate (Sigma, S8636), 50 μ M 2-mercaptoethanol (Life Technologies, 31350), 50 μ M MEM non-essential amino acids (Life Technologies, 11140) and 2.5 μ g/ml insulin (Sigma, I9278), 25 U/ml P/S

Table 3: NSC differentiation Concentrations of supplements freshly added to each MC (indicated by a "+" sign): 10 μ M SB, 1 μ M dorsomorphin, 20 ng/ml FGF-2.

2.1.3.2 Astrocyte differentiation from NSC stage Following NSC differentiation, astrocytes were generated using two differentiation protocols:

1. Astrocyte differentiation using STEMCELL Technologies Kits
2. Serum-driven differentiation method

At the start of both protocols, on around day 22 of NSC differentiation from iPSCs, single-cell accutase-passaging was carried out to passage NSCs onto matrigel-coated plates: first, after removal of the culture medium, 50 μ l/cm² accutase (STEMCELL Technologies, 7920) was added to the cells and incubated at 37 °C for 5 minutes. The cells were then rinsed off and diluted 1:10 in PBS. After centrifugation at 400 g for 5 minutes, the cell pellet was finally resuspended with NMM and plated at 15000 cells/cm². At this point, the day count of differentiation was restarted, with the xth day from the NSC stage henceforth referred to as [Dx].

1. Astrocyte differentiation using STEMCELL Technologies Kits The following procedures to differentiate astrocytes from NSCs are based on the protocols from STEMCELL Technologies (Document #DX20345, Version 1.2.0).

The day after single-cell passaging, [D1], the medium was changed to astrocyte differentiation medium (ADM). The cultures were continuously maintained in ADM with full MCs daily until 80 - 90% confluency, on around [D7], when accutase-passaging was repeated, after which full MCs with ADM were performed every other day. When cultures were 80 - 90% confluent on [D14], the differentiating astrocytes were accutase-passaged once more in ADM. The next day, the medium was switched to astrocyte maturation medium (AMM) and maturing astrocytes were continually maintained with full MCs every other day. Accutase-passaging was performed whenever the cultures reached 90 - 95% confluency. After accutase-passaging in AMM for three times in total, which was usually achieved on around [D50] from the NSC stage, the differentiated cells should have reached the stage of mature astrocytes.

A cheaper method of maintaining the astrocytes by a self-assembled maintenance medium, termed astrocyte maintenance medium (astro main), was also tested for its viability, because the commercially available AMM was more costly. Therefore on around [D50], mature astrocytes were separated into two groups, one half maintained in AMM and the other half in astro main. Details of MCs for astrocyte differentiation using STEMCELL Technologies Kits and composition of astro main are listed in Table 4.

[D0]:	NSCs	
	Passaged NSCs in NMM	
[D1 - 6]:	Full MCs with ADM daily	
[D7]:	Passaged differentiating astrocytes in ADM	1st passage in ADM
[D8 - 13]:	Full MCs with ADM every other day	
[D14]:	Passaged differentiating astrocytes in ADM	2nd passage in ADM
[D15 - 20]:	Full MCs with AMM every other day	
[D21]:	Passaged maturing astrocytes in AMM	1st passage in AMM
[D22 - 27]:	Full MCs with AMM every other day	
[D28]:	Passaged maturing astrocytes in AMM	2nd passage in AMM
[D28 - 34]:	Full MCs with AMM every other day	
[D35]:	Passaged maturing astrocytes in AMM	3rd passage in AMM
[D36 - 42]:	Full MCs with AMM every other day	
[D50]:	Passaged mature astrocytes 50% in AMM and 50% in astro main	
[D50 -]:	Mature astrocytes	
	Full MCs with AMM and astro main respectively every other day	
ADM:	From Astrocyte Differentiation Kit (STEMCELL Technologies, 8540)	
AMM:	From Astrocyte Maturation Kit (STEMCELL Technologies, 8550)	
Astro main:	1:1 mixture of DMEM/F12 GlutaMAX:Neurobasal medium, 1% N2, 1 mM sodium pyruvate, 100 U/ml P/S	

Table 4: Astrocyte differentiation using STEMCELL Technologies Kits For all cell passaging during astrocyte differentiation: accutase was used as the dissociation agent, centrifuged cells were resuspended in the previous original medium followed by switching to the new medium the next day, and plated on matrigel-coated plates, unless otherwise stated.

2. Serum-driven differentiation method The following procedures to differentiate astrocytes from the NSC stage are derived from the method described by Santos et al. (2017).

The day after single-cell accutase-passaging of NSCs, the medium was changed to astrocyte induction medium (AIM) freshly supplemented with 10 ng/ml human leukaemia inhibitory factor (hLIF) (Peprotech, 300-05-5). Full MCs with LIF-containing AIM were performed daily until the next passage. At 90% confluency usually on around [D7], cells were passaged with accutase onto matrigel-coated plates at 15000 cells/cm² in LIF-supplemented AIM. LIF treatment was maintained for 14 days from the NSC stage. The maturing astrocytes were continuously maintained in AIM with the same frequency of MCs. However from [D15] onwards, they were passaged onto to non-coated wells instead of matrigel-coated plates. Mature, functional astrocytes should be attained 3 - 5 weeks from the serum addition at the NSC stage. Details of MCs for astrocyte differentiation with the serum-driven method and composition of AIM are listed in Table 5.

[D0]:	NSCs
	Passaged NSCs in NMM
[D1 - 6]:	Full MCs with AIM + hLIF daily
[D7]:	Passaged differentiating astrocytes in AIM + hLIF
[D8 - 13]:	Full MCs with AIM + hLIF every other day
[D14]:	Passaged maturing astrocytes in AIM + hLIF
[D15 -]:	Full MCs with AIM every other day
	Passaged maturing astrocytes with accutase onto non-coated plates in AIM
[D21 -]:	Mature astrocytes

AIM:	DMEM/F12 GlutaMAX, 10% FBS, 2% B27, 1% N2, 100 U/ml P/S
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Table 5: Serum-driven differentiation method Concentrations of supplements freshly added to each MC (indicated by a "+" sign): 10 ng/ml hLIF. For all cell passaging during astrocyte differentiation: accutase was used as the dissociation agent, centrifuged cells were resuspended in the previous original medium followed by switching to the new medium the next day, and plated on matrigel-coated plates, unless otherwise stated.

An overview of the two tested astrocyte differentiation protocols is showcased in Figure 3.

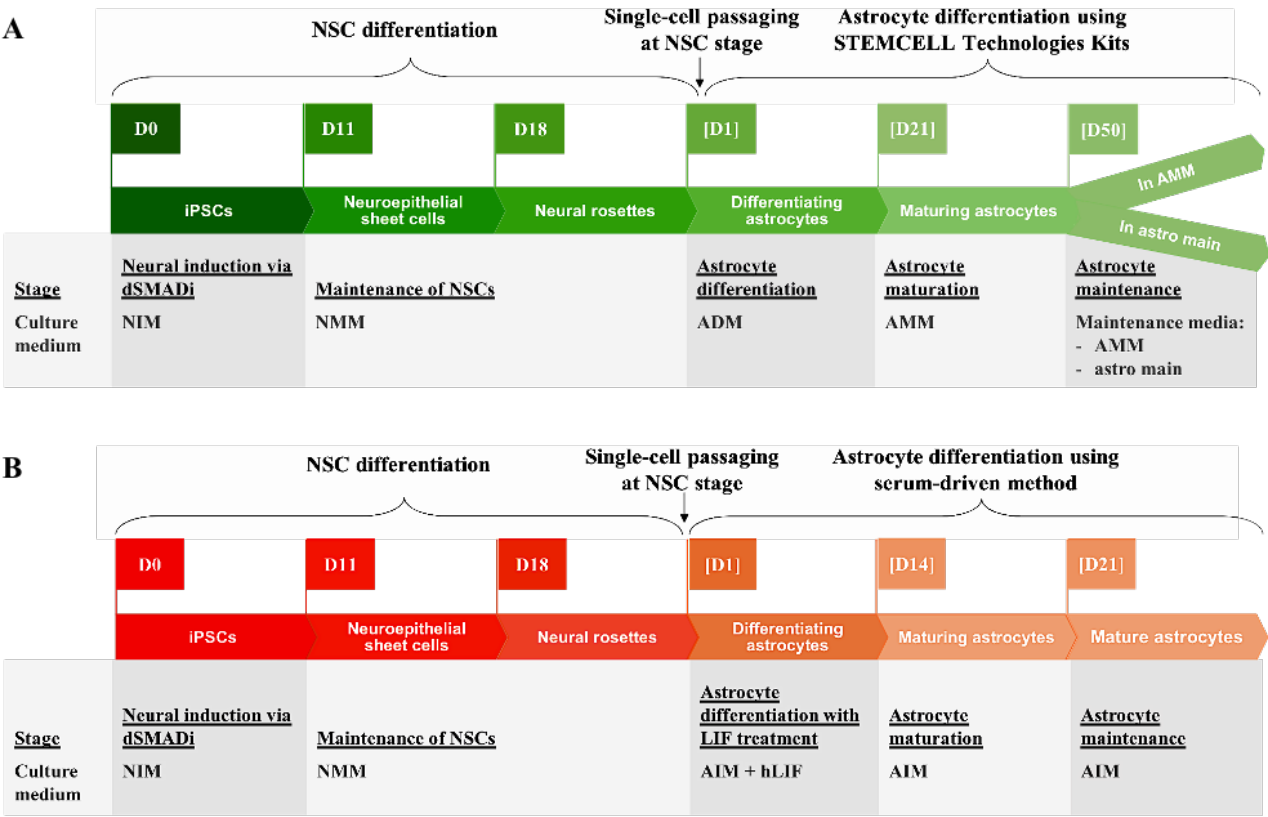


Figure 3: Overview of astrocyte differentiation protocols (A) Astrocyte differentiation protocol using STEMCELL Technologies Kits. (B) Serum-driven differentiation method.

2.1.3.3 Neuronal differentiation Neurons were differentiated using the neurogenin2 (NGN2) differentiation protocol first described in (Y. Zhang et al., 2013). Neurons were differentiated from NSCs that had been previously transduced with a rt-TA3 construct and a tetracycline-inducible NGN2 construct. The presence of a tetracycline such as doxycycline induces NGN2 overexpression by forming a complex with the constitutively expressed rt-TA3 protein that only binds to the promoter of the NGN2 construct when it is tetracycline-bound. These transduced NSCs were plated onto matrigel-coated wells at 25000 cells/cm² and maintained in NSC medium. Neuronal differentiation was induced by fresh addition of doxycycline into each of the daily MCs for the following three days. On day 2 and day 3, puromycin was added to select for the successfully transduced NSCs. On day 1 and 2, neurogenin2 medium 1 (NGN1 medium) was used; on day 3, the medium was switched to neurogenin2 medium 2 (NGN2 medium). From day 3 - 10, differentiating neurons were maintained in NGN2 medium MCs freshly supplemented with different combinations of growth factors on different days (Table 6). From day 10 onward, all MCs switched to using BrainPhys medium without any supplements. Differentiated neurons were considered ready for use by day 12 and were continuously maintained in BrainPhys medium with half MCs ever other day until up to day 21. Details of media changes for the NGN2 differentiation protocol and composition

of NSC medium, NGN1, NGN2, and BrainPhys media are listed in Table 6.

Day 0:	NSCs
	Plated transduced NSCs in NSC medium
Day 1:	Full MC with NGN1 medium + BDNF + NT-3 + doxycycline
Day 2:	Full MC with NGN1 medium + BDNF + NT-3 + doxycycline + puromycin
Day 3:	Full MC with NGN2 medium + BDNF + NT-3 + doxycycline + puromycin
Day 4:	Half MC with NGN2 medium + BDNF + NT-3
Day 6:	Half MC with NGN2 medium + BDNF + NT-3 + Ara-C
Day 8:	Half MC with NGN2 medium + BDNF + NT-3 + Ara-C
Day 10 - 21:	Half MCs with BrainPhys medium every other day
Day 12 - :	Differentiated neurons
NSC medium:	1:1 mixture of DMEM/F12 GlutaMAX:Neurobasal medium, 2% B27 without vitamin A (Invitrogen, 12587010), 1% N2, 50 μ M 2-mercaptoethanol, 100 U/ml P/S + 20 ng/ml BDNF + FGF-2 + EGF
NGN1 medium:	DMEM/F12 GlutaMAX, 1% N2, 10 μ M MEM non-essential amino acids, 100 U/ml P/S
NGN2 medium:	Neurobasal medium, 2% B27, GlutaMAX supplement (Thermo Fisher, 35050061), 100 U/ml P/S
BrainPhys medium:	BrainPhys Medium (STEMCELL Technologies, 05790), 2% B27 without vitamin A, 50 U/ml P/S

Table 6: Neurogenin2 differentiation protocol Concentrations of supplements freshly added to each MC (indicated by a "+" sign), unless otherwise specified: 10 ng/ml BDNF (Peprotech, AF-450-02), 10 ng/ml NT-3 (Peprotech, 450-03), 4 μ g/ml doxycycline (Tocris, 4090), 0.4 μ g/ml puromycin (InvivoGEN, ant-pr-1), 2 μ M Ara-C (Sigma, C1768), 10 ng/ml FGF-2, 10 ng/ml EGF (RD Systems, 236-EG-01M).

2.1.4 Neuron-astrocyte co-culture

Differentiating neurons were maintained in mono-culture until day 12 of the NGN2 protocol when rat primary cortical astrocytes were thawed at a plating density of 15000 cells/cm² in normal rat astrocyte medium. It was switched back to BrainPhys medium as neurons do not tolerate rat astrocyte medium. Co-cultures were maintained in BrainPhys medium until day 21 of the NGN2 protocol.

2.2 Immunofluorescence

Cultures were washed with PBS and fixed using 4% paraformaldehyde (PFA) (Alfa Aesar, J61899) in PBS for 10 minutes, and then permeabilised with 0.1% Triton X-100 (Sigma, T9284-100ML) for 15 minutes. After that, they were pre-incubated with blocking solution, 10% goat serum in PBS (Sigma, G9023) for one hour, followed by incubation with primary antibodies in 5% goat serum overnight. The next day, cells were washed in PBS three times each for 5 minutes before subsequently incubated in secondary antibody in 5% goat serum for 45 minutes. Three 5-minute PBS washes succeeded the incubation, after which cells were incubated with 600 nM 4',6-Diamidino-2-Phenylindole (DAPI) (Thermo Fisher, D1306) for nuclear staining for 10 minutes. Cells were mounted using Fluorescence

Mounting Medium (DAKO, S3023) on microscope slides, and imaged with a confocal microscope (Olympus Fluoview FV1000). See Table 7 for antibody details.

Antibody	Dilution	Species	Company	Catalogue no.
anti-PAX6 (1°)	1:1000	rabbit	BioLegend	PRB-278P
anti-NESTIN (1°)	1:1000	mouse	Abcam	ab22035
anti-NFIA (1°)	1:1000	rabbit	Abcam	ab41851
anti-GFAP (1°)	1:2000	chicken	Abcam	ab4674
anti-S100 β (1°)	1:1000	rabbit	Dako	Z0311
anti-TUJ1 (1°)	1:1000	rabbit	BioLegend	PRB-435P
594 fluorophore-conjugated anti-rabbit (2°)	1:1000	goat	Abcam	ab150113
488 fluorophore-conjugated anti-mouse (2°)	1:1000	goat	Abcam	ab150169
488 fluorophore-conjugated anti-chicken (2°)	1:1000	goat	Abcam	ab150080

Table 7: Primary (1°) and secondary (2°) antibodies used in immunofluorescence All antibodies were diluted in 5% goat serum in PBS.

Blind automated image analysis was performed with Fiji/ImageJ for counting of stained cells. Quantification of staining was visualised in plots of percentages of positively-stained cells out of the total cell count (DAPI-stained) in each IF image.

2.3 qRT-PCR

Sample preparation and RNA purification

Cells were collected using accutase and lysed by adding 300 μ l TRIzol reagent (Invitrogen, 15596026). The total RNA was extracted using the Direct-zol RNA MicroPrep Kit (Zymo Research, R2062). Purified RNA was measured for its concentration and purity using a Nanodrop. RNA samples were stored at -80 °C.

Reverse transcription

Extracted RNA was reverse transcribed into cDNA as follows: a 13 μ l mastermix was prepared for each reaction mixture, consisting of 1.3 μ l random hexamers (Invitrogen, N8080127), 1 μ l 10 mM dNTP mix, 150 ng RNA sample, and DNase-RNase-free water. The reaction mixtures were heated at 65 °C for 5 minutes and left on ice for at least 1 minute. Then, 7 μ l of a different mastermix, consisting of 4 μ l 5x first strand buffer (Thermo Fisher, 18080044), 1 μ l 0.1 M DTT (Thermo Fisher, 18080044), 1 μ l RNase OUT recombinase RNase inhibitor (Thermo Fisher, 10777019), 1 μ l SuperScript III RT (Thermo Fisher, 18080044), was added to each reaction mixture. Reaction mixtures of 20 μ l were incubated at room temperature for 5 minutes, then run in the PCR machine at 50 °C for 45 minutes

and subsequently at 70 °C for 15 minutes. cDNA collected was also measured for its concentration and purity using a Nanodrop, and then diluted to 100 ng/μl for subsequent use in qRT-PCR. cDNA samples were stored at -20 °C.

qRT-PCR procedures

Each reaction mixture consisted of the following: 2 μl 100 ng/μl cDNA template (200ng) for each sample, 10 μl SYBR green mastermix (Thermo Fisher, 4309155), 6.2 μl DNase-RNase-free water, and 0.9 μl each of forward and backward primers for a target gene. Primer sequences are displayed in Table 8.

Target gene	Forward primer	Reverse primer
EAAT1	AGGCAGTAGCAGCTGTGTTT	TGCACAGAATGCTTGGGGTC
EAAT2	GAGGATGGGGGTGACCTAGT	GATGACTGTCTGTCATTTCGC
GFAP	AGAGATCCGCACGCAGTATG	TCTGCAAACCTGGAGCGGTA

Table 8: Primers used in qRT-PCR Primers were custom made from Integrated DNA Technologies.

Gene expression was analysed using the $\Delta\Delta C_t$ method: all results were normalised to expression of housekeeping genes, β -actin and hypoxanthine phosphoribosyltransferase (HPRT), and normalised to iPSCs as the negative control (baseline = 1) to obtain the fold expression.

2.4 Western blotting

Sample preparation

Astrocytes were collected in PBS using accutase and lysed in 100 μl RIPA buffer (Thermo Scientific, 8990) supplemented with protease and phosphatase inhibitor (PPI) (Thermo Scientific, 78441) at 1:100. Protein concentration for each sample was determined using the Micro Protein BCA Assay Kit (Thermo Scientific, 23235) by comparing against a bovine serum albumin (BSA) standard curve constructed according to manufacturer's instructions.

Western blotting procedures

After obtaining the protein-containing lysates, a 15 μl mastermix was prepared, containing 2.5 μl loading dye (Sigma, S3401-1VL), 1 μl NuPage Sample Reducing Agent (Invitrogen, NP0004), 5 μg protein and autoclaved water.

Samples were denatured at 70 °C for 10 minutes and kept on ice afterwards. 5 μl SeeBlue Plus2

Pre-stained Protein Standard (Invitrogen, LC5925) was loaded in the ladder lane and 15 µl of each sample mastermix were loaded onto a 1.5 mm, 15-well NuPAGE Bis-Tris 4-12% gel (Invitrogen, NP0336PK2) in triplicates. Proteins were separated across the gel in MES SDS running buffer (Life Technologies, NP0002) by applying 150 V for 35 minutes.

After gel electrophoresis, proteins were transferred to the nitrocellulose membrane using the TransBlot system (BioRad). The membrane was blocked by incubating it in 5% BSA (Sigma, A2153) in TBST buffer (10% TBS (Biorad, 1706435) in dH₂O + 0.1% Tween20 (Sigma, P1379) for 1 hour at room temperature on a shaker. Then, it was incubated with primary antibodies in 1% BSA for 2 hours at room temperature, followed by three washes in TBST for 5 minutes each. Afterwards, the membrane was incubated in secondary antibody in 1% BSA for 45 minutes at room temperature on a shaker. After the secondary antibody incubation, the membrane was washed three times with TBST for 10 minutes and ECL detection (Pierce ECL Western Blotting Substrate, Thermo Scientific, 32106) was performed according to manufacturer's instructions. Blots were imaged using the AI600 Chemiluminescent Imager (Amersham Imager 600) with 2-second, 10-second, and 1-minute exposures respectively. See Table 9 for antibody details.

Antibody	Dilution	Species	Company	Catalogue no.
anti-phospho-S6 (Ser235/236) (1°)	1:2000	rabbit	Cell Signalling	4858
anti-S6 (1°)	1:2000	rabbit	Cell Signalling	2217
HRP-conjugated anti-β-actin (1°)	1:25000	mouse	Abcam	ab49900
anti-rabbit (2°)	1:5000	goat	Abcam	ab6721

Table 9: Primary (1°) and secondary (2°) antibodies used in western blotting All antibodies were diluted in 1% BSA in TBST.

Imaged blots were analysed with ImageJ to obtain the band intensities of proteins of interest normalised to the β-actin band intensity to generate densitometry data for comparing the protein abundance between samples.

2.5 Glutamate uptake assay

Glutamate uptake and sample collection

Cultures were allowed to equilibrate for 10 minutes in Hank's Buffered Salt Solution (HBSS) (Gibco, 14025). After equilibration, HBSS was replaced by 5 mM L-glutamate (Sigma, G1251) in HBSS. 100 µl glutamate-containing HBSS sample were repeatedly collected for each well over a series of

timepoints (0, 10, 30, 60, 120, 180 minutes). The glutamate samples collected were put on dry ice.

Glutamate assay

Concentration of glutamate in each sample was quantified using the Glutamate Assay Kit (Sigma, MAK004) by comparing against the glutamate standard curve constructed according to manufacturer's instructions. An overall schematic of the workflow of the glutamate uptake assay is displayed in Figure 4.

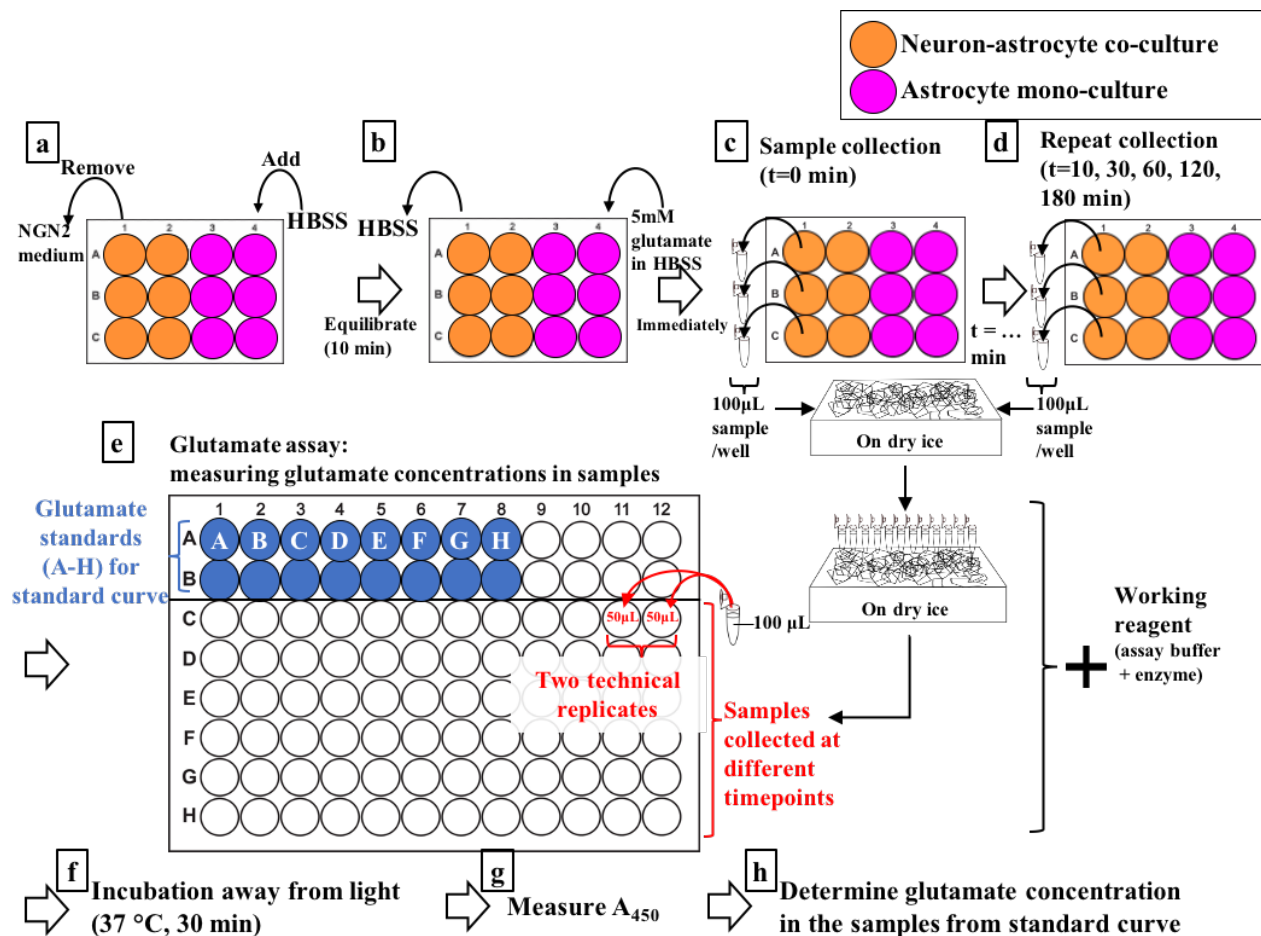


Figure 4: Schematic diagram of glutamate uptake assay Steps: (a) Removal of culture media and addition HBSS for 10 minutes of equilibration. (b) Removal of HBSS and addition of glutamate-containing HBSS. (c) Sample collected at $t=0$ min upon introducing glutamate. (d) Repeated sample collection for the remaining timepoints over the course of glutamate uptake. (e) Mixture of glutamate assay reagents and samples. (f) Incubation at 37°C for 30 minutes for enzymatic reaction to result in the colourimetric (450 nm) product depending on the glutamate concentration in samples. (g) Measurement of absorbance at 450 nm. (h) Determination of the glutamate concentration in the samples from the standard curve.

Normalisation of glutamate uptake to protein concentrations in samples

Proteins were collected by lysis with the aforementioned method using RIPA buffer and PPI, followed by determination of protein concentration in each sample with the BCA Assay Kit. Glutamate concentrations for each sample were to be normalised to the respective protein concentrations.

The change in normalised glutamate concentration in the cultures in each experimental run was plotted against time, to obtain the area under curve (AUC).

2.6 Statistical analysis

One-way ANOVA and the Bonferroni's post-hoc test was used to analyse IF, qRT-PCR and western blotting results. A Student's t-test was carried out on the AUC values from the glutamate uptake assay to test for statistical significance for different cultures. All statistical analyses were performed using GraphPad Prism. P values smaller than 0.05 were considered significant. Data are expressed as mean $\pm$ SEM.

Statistical analyses and post-hoc tests were not performed for data sets where the total sample size for all groups (n) equals the number of groups (k), because theoretically n-k should exceed zero to enable calculation of variance. Nevertheless, sample sizes were all very small in general. Thus, a normality test or a Levene's test, to check for normal distribution and equal variances respectively, might not be meaningful. Hence for all analyses, normality and homoscedasticity of the data were assumed.

Coefficient of variation (CV) was used as a statistical measure to quantify the variation in data series with high variability, calculated by dividing the standard deviation (σ) by the mean (μ):

$$CV = \frac{\sigma}{\mu}$$

2.7 Criteria for comparisons of differentiation protocols

The suitability of astrocyte differentiation protocols to generate mature human astrocytes was compared with regard to the following criteria: (A) cell maturity and (B) maintenance of total cell count, as well as three minor parameters including (I) time, (II) cost, and (III) workload of the protocols.

(A) Cell maturity

The expression levels of four established astroglial markers (EAAT1, EAAT2, GFAP and S100 β) were evaluated to compare the maturity of cells in differentiation, with either IF or qRT-PCR or both techniques (Table 10).

Marker	Technique(s) used to detect expression
EAAT1	qRT-PCR
EAAT2	qRT-PCR
GFAP	IF and qRT-PCR
S100 β	IF

Table 10: Astroglial markers and techniques used for detection to identify astrocyte maturity

IF captured the staining of cultures at a series of timepoints in IF images, allowing visualisation of general expression pattern of GFAP and S100 β during astrocyte differentiation, normalised to the staining to the total number of cells. qRT-PCR measured the expression level normalised to housekeeping genes as well as iPSCs to obtain the fold expression for cells generated at the end of a differentiation. The expression or presence of an astroglial marker was considered to be genuine if its fold expression was significantly higher than the iPSC baseline (= 1) with significance at 0.05 level.

(B) Maintenance of total cell count

The total cell count was indicated by the IF staining of blue fluorescent DAPI, which stains the DNA of fixed cells. The maintenance of total cell count is reflected by its change over the course of astrocyte differentiation.

(I) Time

The time was defined by the minimum number of days required for the generation of mature astrocytes in each protocol.

(II) Cost

The cost of each protocol was calculated from the price of the kits or reagents needed to complete one round of differentiation maintaining cultures of a full 6-well plate.

(III) Workload

The workload for each protocol was compared in terms of minimum number of passages required before astrocytes reach a mature developmental stage.

3 Results

3.1 Suitability of astrocyte differentiation protocols

SBAD-03 iPSCs were differentiated into astrocytes with the two protocols:

1. Astrocyte differentiation using STEMCELL Technologies Kits
2. Serum-driven differentiation method

The following are the differentiation outcome from each of the protocols, for comparison of their suitability in generating mature human astrocytes in the discussion section.

1. Astrocyte differentiation using STEMCELL Technologies Kits

Two differentiation rounds have been conducted using this protocol.

(A) Cell maturity

IF GFAP expression peaked on [D37] with $5.2\% \pm 1.6\%$ in the first round, and on [D27] with $10.9\% \pm 6.3\%$ in the second round (Fig 5Ai,ii); whereas S100 β expression peaked on [D14] with $19.1\% \pm 3.7\%$ in the first round, and on [D27] with $11.0\% \pm 3.0\%$ in the second round (Fig 5Bi,ii). This suggests an overall decline of GFAP and S100 β expression over time during astrocyte differentiation. Figure 6A shows the general change in GFAP and S100 β expression in the first round of differentiation with this protocol by compiling a representative IF image for each time point.

For both rounds of differentiation, there were no significant differences in the percentages of both GFAP⁺ and S100 β ⁺ cells between the two final media, AMM and astro main (GFAP⁺: first round: AMM: $1.0\% \pm 0.3\%$, astro main: $0.5\% \pm 0.3\%$, $P > 0.9999$; second round: AMM: $0.0\% \pm 0.0\%$, astro main: $0.6\% \pm 0.6\%$, $P > 0.9999$) (S100 β ⁺: first round AMM: $1.7\% \pm 1.0\%$, astro main: 1.2% , $P > 0.9999$; second round: AMM: 0.0% , astro main: 6.0% , $P > 0.9999$). However, astrocytes cultured in AMM and astro main had a different morphological appearance. Differentiation cultures switched to culturing in astro main showed a spikier appearance than those maintained in AMM. Representative images are shown in Figure 6C.

qRT-PCR Cells showed EAAT1 expression in both differentiation rounds when using AMM as the final medium (first round: 22.0, $P < 0.0001$; second round: 8.6 ± 0.6 , $P = 0.0038$). Nonetheless, the fold expression were significantly different between the two rounds ($P = 0.0001$). On the other hand,

cells in astro main showed EAAT1 expression in only the second round (first round: 1.5, $P > 0.999$; second round: 9.4 ± 0.6 , $P = 0.0022$), additionally evidenced by the significant difference comparing the two rounds with each other ($P = 0.0030$) (Fig 7Ai).

Most samples showed undetectable levels of EAAT2 indicating very low levels of EAAT2 expression (Fig 7Aii).

As for GFAP, only cells in astro main in the first round showed significantly different expression from the iPSC baseline (first round: AMM: 6.0, $P > 0.999$, astro main: 24.1, $P = 0.0364$; second round: AMM: 5.5 ± 2.5 , $P > 0.999$, astro main: 1.5 ± 0.7 , $P > 0.9999$). This is also evidenced by the significant difference in the pairwise comparison for cultures in astro main between the two rounds ($P = 0.0058$) (Fig 7Aiii).

(B) Maintenance of total cell count

The total cell count remained relatively stable in both rounds. When comparing between the highest and the lowest staining percentages, there was a 65% and 54.8% decrease in the first and second round respectively (first round: 410.0 ± 106.6 on [D37] to 124.7 ± 45.8 on [D62] in astro main; second round: 287.4 ± 26.0 on [D27] to 145.5 ± 23.9 on [D64]) (Fig 5Ci,ii).

2. Serum-driven differentiation method

Only one round of differentiation could be conducted with this protocol.

(A) Cell maturity

IF Expression of both astrocytic markers, GFAP and S100 β peaked on [D44], with $9.7\% \pm 2.2\%$ GFAP⁺ cells (Fig 5Aiii), and 25% S100 β ⁺ cells (Fig 5Biii).

qRT-PCR Fold expression of EAAT1 and GFAP showed no significant differences from the iPSC baseline (EAAT1: 1.0 ± 0.2 , $P > 0.9999$; GFAP: 1.2 ± 0.5 , $P > 0.9999$) (Fig 7Ai, iii), whereas EAAT2 expression level was undetectable (Fig 7Aii).

(B) Maintenance of total cell count

The total cell count showed a clear decline over time, with a percentage drop of 80.3% and 98.9%, from 306.0 ± 31.6 on [D30] to 60.2 ± 8.3 [D44] and 3.3 ± 3.3 on [D100] respectively (Fig 5Ciii). The decrease in total number of cells was also evident in microscopic IF images (Fig 6B).

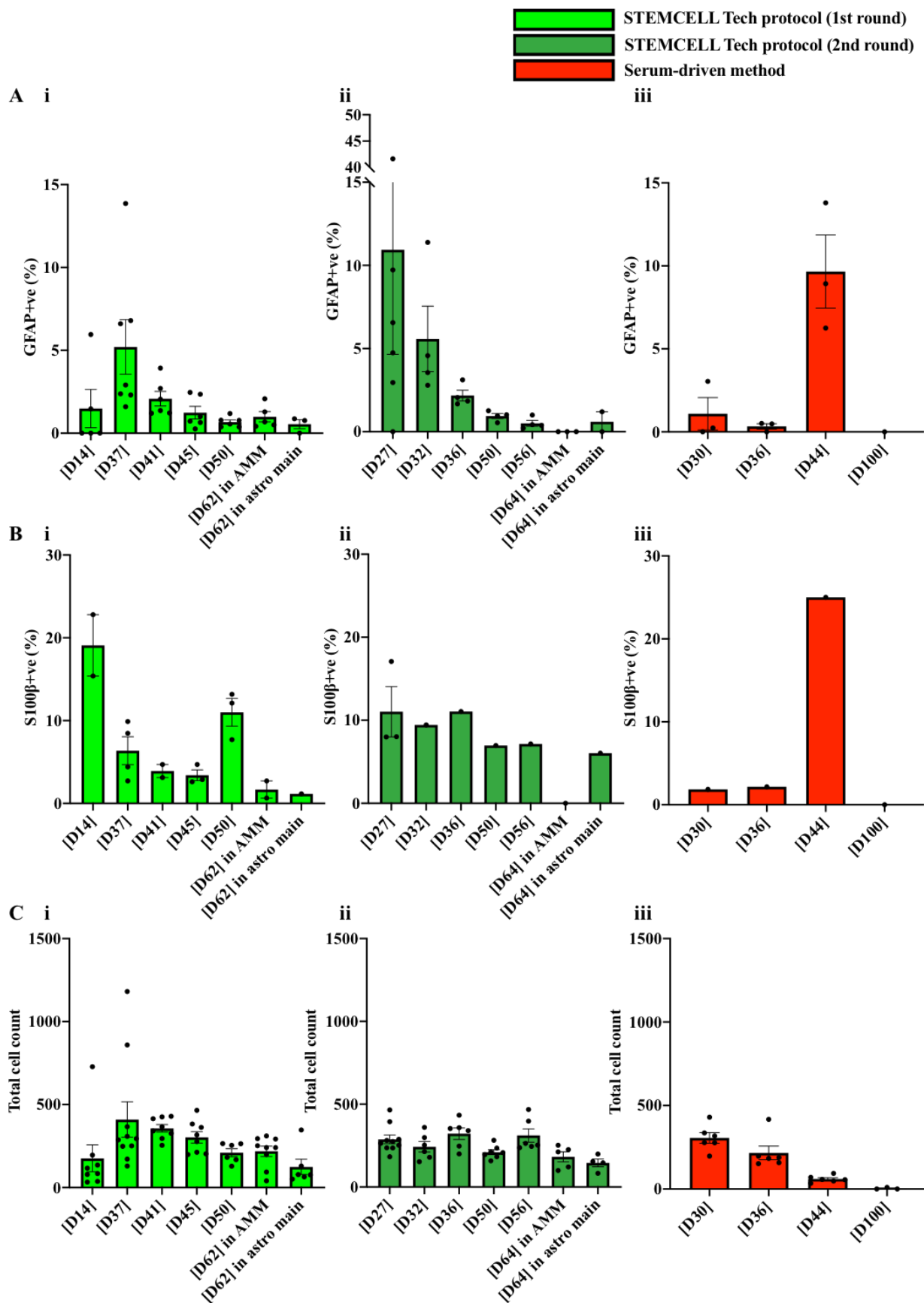


Figure 5: Quantification of immunofluorescence results (A) percentages of GFAP⁺ cells and (B) S100β⁺ cells, and (C) total number of cells (DAPI-stained) generated by astrocyte differentiation using STEMCELL Technologies Kits in the first round (light green) and second round (dark green), and by the serum-driven differentiation method (red). Each data point corresponds to positive staining in each captured image. Data are expressed as mean ± SEM.

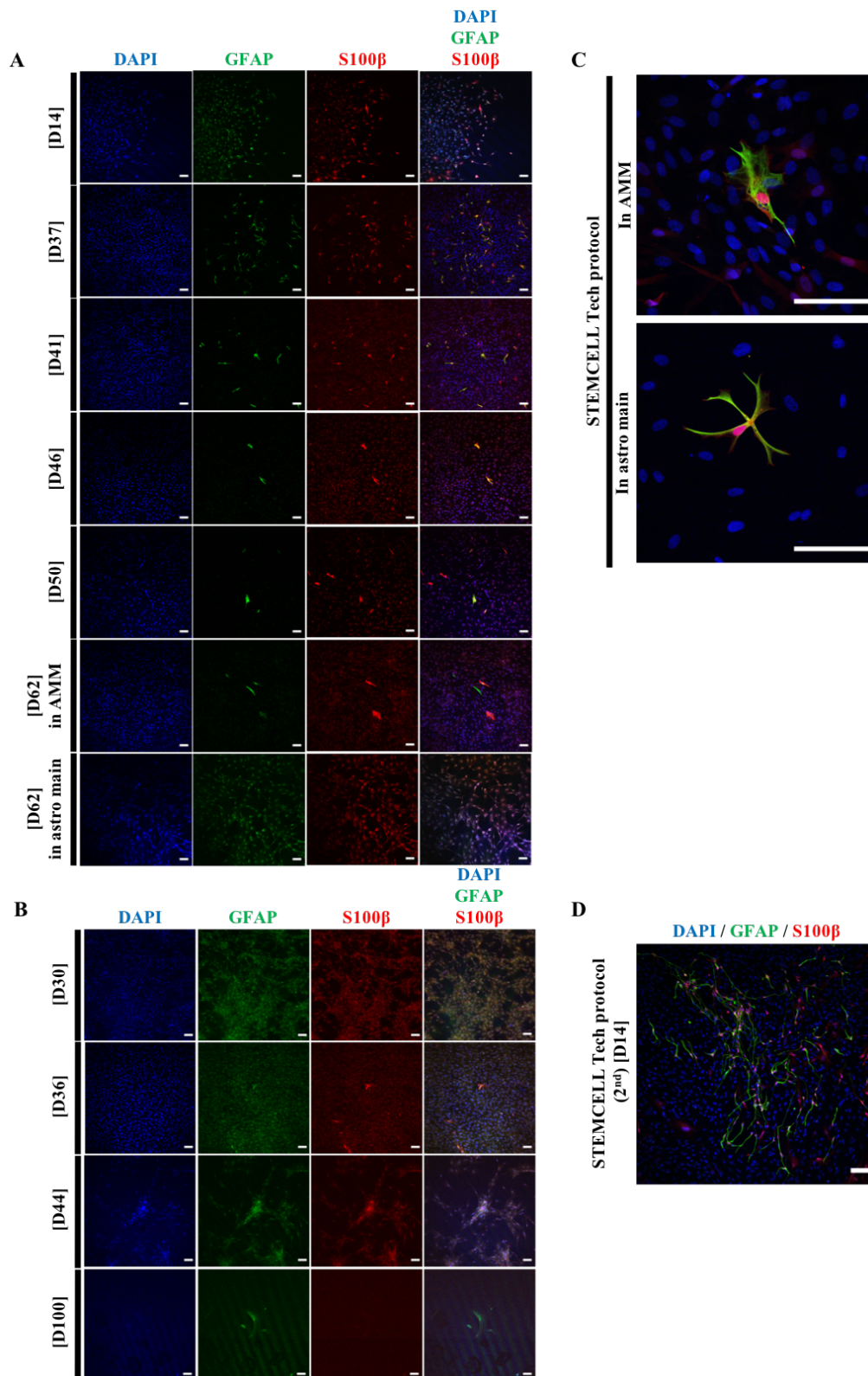


Figure 6: Immunofluorescence images Representative IF images taken at different timepoints during astrocyte differentiation (A) using STEMCELL Technologies Kits (first round), and (B) with the serum-driven method, showing the staining of DAPI (blue), GFAP (green), and S100β (red). High auto-fluorescence was observed in IF images with few to zero stained cells, particularly for the serum-driven method. (C) Representative IF images of higher magnification showing morphology of GFAP⁺ astrocytes cultured in different final media (AMM vs astro main) at the end stage of differentiation for the protocol using STEMCELL Technologies Kits. (D) Representative composite IF image showing a population of clustered GFAP⁺ cells in differentiation using STEMCELL Technologies Kits. Scale bars represent 100 μm.

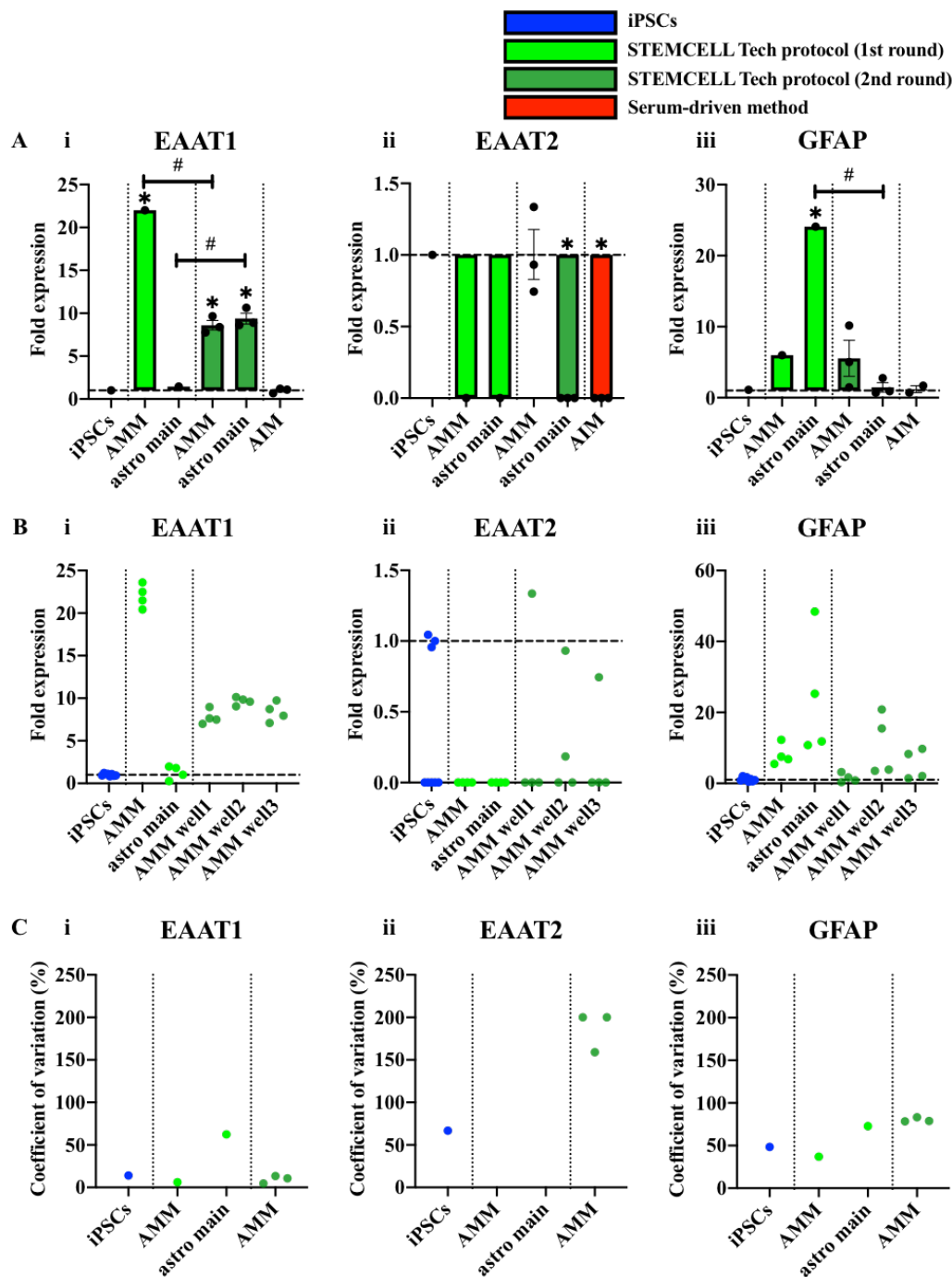


Figure 7: qRT-PCR results (A) Fold expression of EAAT1, EAAT2, and GFAP in cells generated from astrocyte differentiation using STEMCELL Technologies Kits in the first (light green) and second (dark green) rounds, cultured in AMM (first round: n=1; second round: n=3) and astro main (first round: n=1; second round: n=3), as well as by the serum-driven method (red), cultured in AIM (n=1). Cells were at their final differentiation timepoints: [D62] and [D64] for the first and second rounds of astrocyte differentiation using STEMCELL Technologies Kits respectively, and [D100] for the serum-driven differentiation method. All expression levels were normalised to housekeeping genes. All samples were compared to iPSCs (blue), with the baseline of fold expression = 1 indicated by the horizontal dashed line. One-way ANOVA with Bonferroni's post-hoc test was used for analyses. * indicates that mean is significantly different from the iPSCs at the 0.05 level. # shows the inter-round difference with significance at the 0.05 level for the same final medium in samples differentiated using STEMCELL Technologies Kits. Data are expressed as mean $\pm$ SEM. (B) Fold expression of technical replicates for samples on which multiple qRT-PCR measurements were performed. (C) Quantification of data variation in (B) by calculating the coefficient of variation (CV).

Data variability

To determine whether inter-round variability in qRT-PCR results arose from practical techniques or a true biological origin, the coefficient of variation (CV) was used to show the degree of data dispersion with respect to the mean in the data series.

Raw data from qRT-PCR were further analysed to identify the source of data variability (Fig 7B,C). Figure 7B displays The individual fold expression of each technical replicate for samples where two separate rounds of qRT-PCR measurements had been performed. Two technical replicates per round were included, resulting in four data points per sample. Figure 7C shows the grouped values of CV percentages calculated for each of the data series reanalysed in Figure 7B.

The data points for EAAT1 are more tightly clustered and less scattered than GFAP. These correspond to the generally higher CV percentages for GFAP than EAAT1 (Fig 7Ci,iii). Thus there is higher variability between replicates in GFAP than EAAT1 expression.

Minor parameters

Apart from considering cell maturity and total cell count for determining the suitability of the protocols, three other criteria were also compared. These were time, cost and workload.

(I) Time

The STEMCELL Technologies protocol mandates that differentiation over a minimum of 50 days from the NSC stage is required before mature astrocytes are generated. This is more than double the time required for the serum-driven method according to Santos et al. (2017), which is 21 days post NSC stage.

(II) Cost

Running one differentiation round with the STEMCELL Technologies protocol requires ~£342, which is around five times more expensive than the serum-driven differentiation method requiring ~£68.

(III) Workload

The STEMCELL Technologies protocol requires at least 6 passages whereas the serum-driven method requires at least 3 passages in total, hence the former is more laborious than the latter (Tables 4, 5).

All protocol comparisons have been summarised in Figure 8.

Differentiation method				1. STEMCELL Tech protocol (1 st (left) and 2 nd (right) rounds)		2. Serum-driven method
Feature						
Generated astrocytes	IF	Maturity (peak % staining)	GFAP	5.2% [D37]	10.9% [D27]	9.7% [D44]
			S100β	19.1% [D14]	11.0% [D27]	25% [D44]
		Morphology		Spikier appearance in astro main than AMM		Fibroblast-like
	qRT-PCR	Maturity at final stage (normalised, mean fold expression) #	EAAT1	AMM: 22.0 Astro main: ns	AMM: 8.6 Astro main: 9.4	ns
			EAAT2	AMM: null Astro main: null	AMM: ns Astro main: null	null
			GFAP	AMM: ns Astro main: 24.1	AMM: ns Astro main: ns	ns
Total cell population	No. of cells at final stage		AMM: 219.4 Astro main: 124.7	AMM: 182.2 Astro main: 145.5	3.3	
	% drop of cell count (highest mean → lowest mean)		356.9 [D41] → 124.7 [D62 in astro main] (~65%)	322 [D36] → 145.5 [D64 in astro main] (~54.8%)	306 [D30] → 60.2 [D44] (~80.3%) → 3.3 [D100] (~98.9%)	
General	Time		After [D50]		After [D21]	
	Cost		More expensive (~£342)		Cheaper (~£68)	
	Workload		More laborious (more passages)		Less laborious (less passages)	

Figure 8: Summary of comparisons between differentiation protocols The two astrocyte differentiation protocols (1. Using STEMCELL Technologies Kits and 2. Serum-driven method) were compared regarding the astrocytes generated in terms of (A) cell maturity, regarding the total cell population in terms of (B) the maintenance of its cell count, as well as regarding (I-III) general features of the protocols (time, cost, and workload). #: Only the values of normalised fold expressions with significance are included; "ns" refers to fold expression with no significant difference from negative control; "null" refers to undetectable expression levels.

3.2 mTOR pathway activation in astrocytes in response to absence of glucose and serum

To optimise assays for examining effects of nutrient supply on mTOR pathway, changes in pS6:S6 ratios in rat astrocytes after loss of nutrients were studied.

Three groups of primary rat astrocytes were cultured in rat astrocyte medium. When 80% confluency was reached, the medium was changed to glucose- and serum-free media for two of the groups respectively, for 24- and 72-hour incubation periods. The two groups were compared against the third group that were continuously cultured in normal rat astrocyte medium (control) to assess the effect of glucose and serum starvation on mTOR pathway activation.

pS6 and S6 bands from western blots were cropped out and shown in Figure 9A, above their corresponding β -actin bands to which each band intensity was normalised. All S6 bands for serum-free condition were generally fainter compared to other bands, evidenced by the supplementary analysis of S6 band intensities in Figure 9C.

When compared to the control, no significant differences in pS6:S6 ratios were observed for rat astrocytes cultured in either glucose-free or serum-free media for both incubation periods (24 hours: control: 0.7 ± 0.1 ; glucose-free: 0.5 ± 0.1 , $P=0.8728$; serum-free: 0.3 ± 0.1 , $P=0.1237$; 72 hours: control: 1.2 ± 0.3 ; glucose-free: 1.5 ± 0.2 , $P>0.9999$; serum-free: 0.6 ± 0.1 , $P=0.4698$) (Fig 9B).

Expression levels of S6 were significantly lower in serum-free medium when compared to normal and glucose-free media respectively for both incubation periods (24 hours: control: 12323 ± 2143 ; glucose-free: 13003 ± 1915 ; serum-free: 2356 ± 383.2 , versus control: $P=0.0262$, versus glucose-free: $P=0.0390$; 72 hours: control: 11673 ± 849.5 ; glucose-free: 13683 ± 382.6 ; serum-free: 1495 ± 110.2 , versus control: $P<0.0001$, versus glucose-free: $P<0.0001$) (Fig 9C).

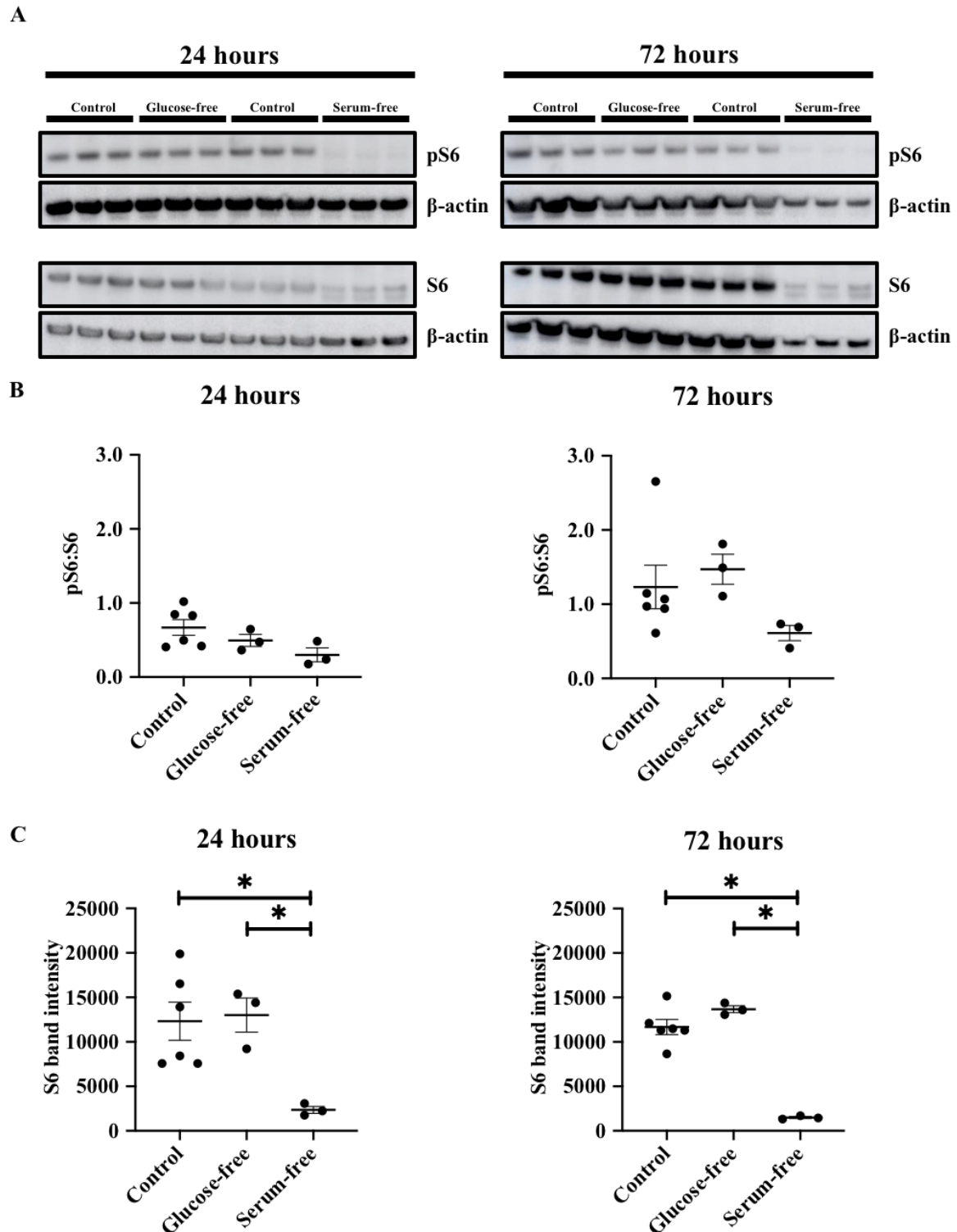


Figure 9: mTOR pathway activation in astrocytes in response to nutrient supply Rat astrocytes were incubated in 3 conditions of rat astrocyte media for 24 and 72 hours: normal (control) (n=6), glucose-free (n=3), and serum-free (n=3). (A) Western blots showing pS6 and S6 bands with the respective β -actin bands for normalisation for each sample. (B) pS6:S6 ratios in rat astrocytes in the different culture conditions from densitometry data. (C) S6 expression levels in rat astrocytes in the different culture conditions. One-way ANOVA with Bonferroni's post-hoc test was used for all analyses. * indicates that mean difference is significant at the 0.05 level. S6 expression (24 hours): $P=0.0262$ (control vs serum-free), $P=0.0390$ (glucose-free vs serum-free); S6 expression (72 hours): $P<0.0001$ (control vs serum-free), $P<0.0001$ (glucose-free vs serum-free). Each data point corresponds to sample collected from each well. Data are expressed as mean $\pm$ SEM.

3.3 Glutamate uptake in neuron-astrocyte co-cultures and astrocyte mono-cultures

To optimise assays for studying astrocytic glutamate uptake function, glutamate uptake assay was carried out on rat primary astrocytes co-cultured with neurons differentiated from the NGN2 protocol, and on astrocyte mono-cultures to investigate the effect of neuronal co-culture on glial function.

For consistency, rat astrocytes thawed in normal rat astrocyte medium were switched to be maintained in BrainPhys medium the day after plating, so that both co-cultures and mono-cultures were kept in BrainPhys medium.

Glutamate concentration consistently decreased over time for both co-cultures and mono-cultures (Fig 10A). Neuron-astrocyte co-cultures displayed a more rapid drop in extracellular glutamate levels than astrocyte mono-cultures, as evidenced by the significant difference in their mean AUC values obtained from the graph of changes in extracellular glutamate levels plotted against time (co-culture: 1900 ± 307.8 , mono-culture: 3240 ± 370.2 , $P=0.0497$) (Fig 10B).

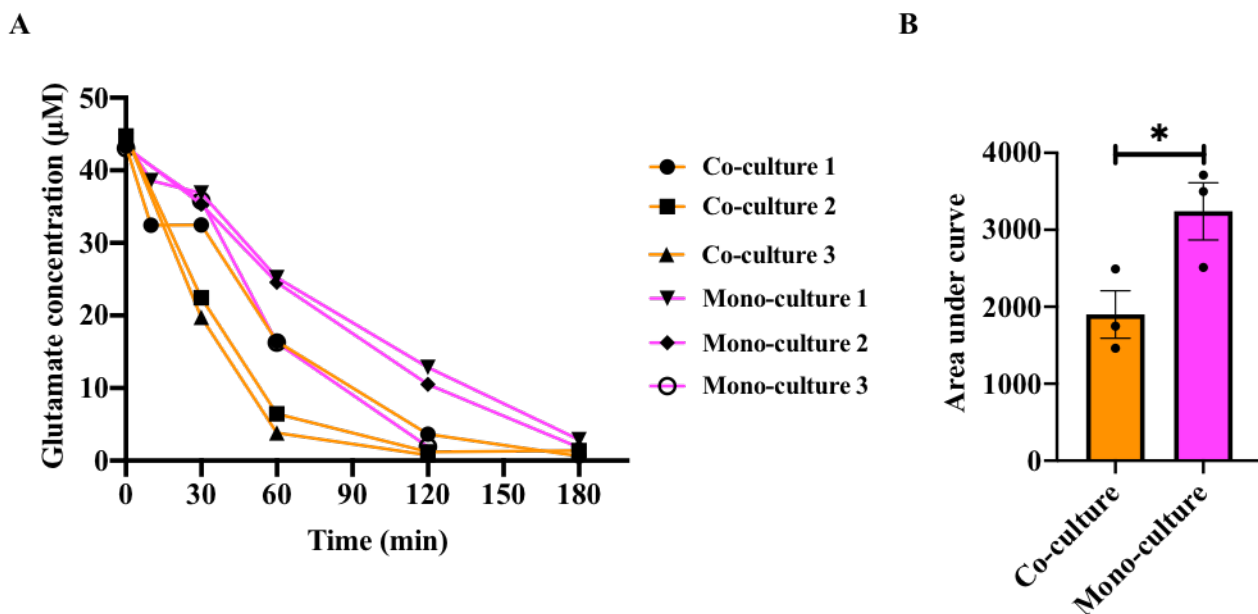


Figure 10: Glutamate uptake in neuron-astrocyte co-culture and astrocyte mono-culture (A) Changes in extracellular glutamate levels over time in each of the co-cultures (orange) and mono-cultures (pink) for each experimental run (n=3). (B) Comparison of area under curve (AUC) values between co-culture and mono-culture for each experimental run obtained from (A). Two-tailed unpaired Student's t-test indicated significant difference at 0.05 level in the mean AUC between co-culture and mono-culture conditions ($P=0.0497$). Each data point corresponds to the AUC obtained from each condition per experimental run. Data are expressed as mean $\pm$ SEM.

4 Discussion

4.1 Comparison of suitability of astrocyte differentiation protocols

On the whole, neither the astrocyte differentiation using STEMCELL Technologies Kits nor the serum-driven differentiation method showed promising results regarding their suitability to generate mature human astrocytes.

In terms of (A) cell maturity, IF and qRT-PCR results showed relatively higher expression of astroglial markers in cells generated by the STEMCELL Technologies protocol than the serum-driven method. Moreover, STEMCELL Technologies protocol also seemed to be comparatively more stable with regard to (B) maintenance of total cell count.

However, a conclusion of STEMCELL Technologies protocol being more suitable is questionable due to two reasons: regarding (A) cell maturity, the GFAP and S100 β staining percentages for both protocols were low compared to the manufacturer's protocol from IF results; secondly, with respect to (B) maintenance of total cell count, the differentiation with the serum-driven method was inadvertently protracted, which was potentially the cause for the large decline in total cell count and the very low number of cells for qRT-PCR. Furthermore, due to coronavirus disease 2019 (COVID-19), a second differentiation round could not be carried out for the serum-driven method to address these uncertainties.

Astrocyte differentiation using STEMCELL Technology Kits is more suitable

(A) Cell maturity

IF Although serum-driven differentiation method seemingly produced a higher peak staining for both GFAP and S100 β than the STEMCELL Technologies protocol, the occurrence of peak staining is probably caused by the low total cell count on [D44] (Fig 5Ciii). Further evidenced by IF images, this implicates that peak staining percentage was a facade of having a high number of positively-stained cells, which was in fact caused by the low total cell count. Representative IF images showed the sparser cell distribution and the low to zero staining on [D44] in Figure 6B. The percentages of GFAP and S100 β staining were all lower for the serum-driven method than the STEMCELL Technologies protocol after addressing the exception on [D44].

qRT-PCR At first glance, qRT-PCR results were complex and difficult to summarise for deducing the stage of maturity of differentiated cultures, because there was little to no inter-round consistency in expression patterns for the same final medium and marker, apart from the consistent negative results for the serum-driven method.

Nonetheless, after summarising the qRT-PCR in Table 11, the confusion seems to be caused by the inconsistency across differentiation rounds, which is not unprecedented. A multi-site omics study reported high inter-round variability in molecular phenotypes in long-term differentiation cultures despite originating from the same iPSC lines. It was indicated that factors leading to variation include passage numbers and progenitor storage (Volpato et al., 2018). Hence, taking into consideration the inter-round variability, it can be deduced that three of the samples generated from the STEMCELL Technologies protocol reached the stage of early astrocytes – cultures in AMM for the first and second rounds, and cultures in astro main for the second round only. As for the cultures in astro main for the first round, it is likely that the GFAP expression indicated the reaching of the stage of early GPCs, rather than late astrocytes. This is because there was no co-expression of EAAT2, an astrocytic marker denoting the most mature developmental stage out of all markers.

Marker	Stage of presentation	iPSCs	1st round		2nd round		AIM
			AMM	astro main	AMM	astro main	
EAAT1	Early astrocytes	-	+++	-	+	+	-
EAAT2	Late postnatal astrocytes	-	-	-	-	-	-
GFAP	Early GPCs, late astrocytes	-	-	+++	-	-	-

Table 11: Summary of astroglial marker expression in samples from qRT-PCR findings

Negligibly low EAAT2 expression for all samples from both protocols indicated that all differentiated cultures have not reached a mature developmental stage. The prevalence of undetectable EAAT2 expression levels (fold expression = 0), for multiple samples could also indicate problematic primers. But this could not be confirmed due to the lack of a positive control for EAAT2 detection.

(B) Maintenance of total cell count

With respect to the total cell count, STEMCELL Technologies protocol is clearly more stable in its maintenance compared to the serum-driven method. The maximum percentage drops were 65% and 54.8% for both rounds of differentiation using the STEMCELL Technologies Kits, whereas the serum-driven method showed a 98.9% drop in total cell count, which was almost a loss of the entire cell population (Fig 5C).

At this point, the protocol using STEMCELL Technologies Kits may appear to be more suitable in generating astrocytes, in terms of the cell maturity and maintenance of total cell count. Nonetheless, this is questionable due to the following explanation.

Explanation for proposal that neither protocol produced promising differentiation outcome

(A) Cell maturity

IF When considering the differentiation using STEMCELL Technologies Kits on its own, it is not an effective strategy to generate mature astrocytes either. IF staining did not increase over the course of astrocyte differentiation but peaked with less than 15% GFAP⁺ and less than 20% S100 β ⁺ cells early in astrocyte differentiation for both rounds, followed by a gradual decrease in subsequent timepoints (Fig 5A-B,i-ii). In contrast, according to the manufacturer's protocol, it is expected that more than 85% GFAP⁺ astrocytes should be attained by the end of the differentiation. In addition, the low percentages of GFAP⁺ and S100 β ⁺ cells might have also affected the detection of any significant differences between the use of AMM or astro main as the final medium.

Indeed, it was suspected that the kits were probably problematic when the study was being carried out correlated to the cessation of their sales. Previous work in the laboratory and neighbour laboratories used the same protocol and successfully generated 70-80% GFAP⁺ cells, reported to be anticipated in the manufacturer's protocol as well. STEMCELL Technologies later ceased the production of the kits and they are currently completely withdrawn from sale. It is not unreasonable to suspect that there was a problematic component in the original formula during its usage resulting in the failure of astrocyte differentiation in the scope of this project.

Hence, the STEMCELL Technologies protocol did not produce high percentages of GFAP⁺ and S100 β ⁺ astrocytes either.

(B) Maintenance of total cell count

The serum-driven method appeared to be less stable in maintaining the total cell count and displayed no expression for any astroglial markers in qRT-PCR. On [D100], the total cell count actually dropped to near zero (3.3 ± 3.3), corresponding to the lack of all three marker expression tested in qRT-PCR. Nevertheless, it could be suspected that this notable decline in total cell count was caused by the inadvertently protracted differentiation process. As a matter of fact, the differentiation cultures were maintained much longer than what was required (21 days from the NSC stage) to obtain mature

astrocytes according to the original protocol (Santos et al., 2017) (Table 5, Fig 8). The main reason for keeping the culture till [D100] was due to microscopic observation of little staining in the IF images on earlier timepoints, with hopeful expectations that the absence of marker staining only indicated the need of more time for cultures to mature, although this was ultimately proven not to be the case. Hence, there was an overall delay in comparison to the original protocol, resulting in these relatively late timepoints for capturing IF images. In hindsight however, it was probably more preferable to have carried out IF on the sparse cultures regardless. Unfortunately, the reasons and uncertainties regarding the failure of the serum-driven differentiation could not be addressed due to the inability to repeat the differentiation by the serum-driven method because of COVID-19. Therefore, these are recommended for future work.

Minor parameters

Furthermore, regarding other minor parameters, the STEMCELL Technologies protocol is more expensive, requiring more time and workload for generating mature astrocytes than the serum-driven method.

Overall, there were no promising results indicating the successful generation of a cell population with a high percentage of mature astrocytes from either protocol in the scope of this project.

Potential improvements

Further rounds of differentiation for the serum-driven method with earlier timepoints for capturing IF images should be undertaken. Other astrocyte differentiation protocols from the list compiled by Chandrasekaran et al. (2016) could also be attempted to establish an effective protocol.

High variability in differentiation data

Apart from the inter-round variability observed, there were also high intra-sample variability in IF and qRT-PCR.

IF Cells were not consistently stained for GFAP, as populations of clustered GFAP⁺ and GFAP⁻ cells were observed. For quantified IF results, there were notably wider dispersion of staining percentages in earlier timepoints, which could be accounted for by the clustered appearance of stained cells, validated by the microscopic observation. This is because each image captures a confined area with a variable number of staining clusters. For instance, the highest staining percentage of 41.6% on [D27] in Figure 5Aii corresponds to one region with particularly high number of clusters of stained cells captured in the IF composite image in Figure 6D.

One way to address this variability is by capturing a consistent and higher number of IF images to confirm that the source of the variability arose from a true biological origin. This was unable to be executed however due to the limited use of microscopy facility during COVID-19.

qRT-PCR In qRT-PCR data, there was an overall higher degree of variability shown in GFAP expression compared to EAAT1, evidenced by the wider dispersion of data points and generally higher CV percentages (Fig 7B,C). Considering there was high intra-sample variability only for GFAP measurements, it is unlikely that technique is the main source of variation. This also narrows down the possible causes for such variability, namely problematic primers, although not confirmed due to the absence of a positive control. It is worth noting that as $CV = \frac{\sigma}{\mu}$, an absolute value of a mean below 1 would result in a CV percentage over 100%. This would account for several CV percentages for EAAT2 expression that were over 100% (Fig 7Cii), which correspond to the respective fold expressions with mean values smaller than 1 due to a number of undetectable measurements (Fig 7Bii).

In addition to the high variability for GFAP expression in qRT-PCR, samples were not consistently prepared in triplicates, which is a limitation. Hence, interpretations from results in Figure 7Aiii warrant more caution.

4.2 mTOR pathway activation in astrocytes in response to loss of nutrient supply

Glucose and serum were removed from normal rat astrocyte medium for respective groups over 24- and 72-hour periods to examine the response of mTORC1 activation to nutrient starvation. Based on increased activation of mTOR pathway upon addition of serum and growth factors (Haselbacher et al., 1979; Thomas et al., 1982), it was expected that the removal of glucose and serum, being a metabolic stress as well as loss of nutrient supply, would result in a decrease in mTOR pathway activation. However, surprisingly, activation of mTORC1 did not significantly decrease in both glucose- and serum-starved rat primary astrocytes at 24 and 72 hours of incubation. On the other hand, the S6 expression in serum-starved rat astrocytes was low after both incubation periods.

Rat astrocytes may potentially require more time for exposure to the nutrient-free culture media, for changes in pS6:S6 to lead to changes in mTOR modulation to manifest. Nonetheless, it would be challenging to attempt a longer incubation time than 72 hours, particularly in the serum-starved cultures, because serum starvation induces growth arrest and increases cell detachment (Hasan et al., 1999). There was an indication of higher cell death in only serum-deprived cultures at 72 hours, with little recovery from the cell pellet in this condition and the resulting relatively faint β -actin bands on the imaged blots (Fig 9A). The mTOR pathway tightly regulates cell survival and proliferation. Hence, modification to the mTOR pathway could result in lower cell turnover and increased apoptosis, which could be a reason for decreased cell survival in the serum-starved cultures.

One way of addressing this is using plates with larger well surfaces, because this would allow more cells to be collected per condition with the same seeding density, providing a wider window for cell death from serum or general nutrient starvation, so that a sufficient number of cells would still be alive after longer incubation periods. Alternatively, one could further incubate the cultures with MHY1485, a compound known to increase mTORC1 activity, to activate the mTOR pathway to enhance cell proliferation. This would further confirm that mTOR modification was indeed the cause of the increased cell death in nutrient-free media.

Low expression levels of the dephosphorylated S6 protein for serum-free cultures in particular indicates that 24 hours of serum starvation is sufficient to significantly reduce its expression, implicating that the presence of serum is necessary for normal S6 expression. The impaired normal S6 expression may have caused or may be associated with the increased cell death in serum-starved cultures, given that S6 is a key signalling molecule downstream of mTORC1 as well as the

knowledge of the significance of mTOR pathway in the regulation of growth and cell survival processes (Feng et al., 2005; Portocarrero et al., 2018). Nonetheless, it is not certain whether both the reduced S6 expression and cell death are mTOR-dependent, without addition of an mTOR activator (MHY1485) to test for the restoration of the phenotype, which is a caveat of the experiment in general: due to the lack of a follow-up addition of mTOR activator or suppressor, it cannot be confirmed whether effects are mainly caused by changes in mTOR pathway activation specifically, or other related mTOR-independent signalling pathways that are also receptive to changes in nutrient supply. For example, previous studies demonstrated that serum could activate RSK that also phosphorylates the downstream S6 via the Ras/ERK signaling cascade independent of mTOR pathway (Roux et al., 2007).

Potential improvements

Overall, rat astrocytes should be exposed to the glucose- and serum-free media for a longer time period to confirm the lack of change in mTOR pathway activation. Regarding serum-starved cultures, a larger starting culture could be considered; the addition of MHY1485 should also be conducted to assess if this mTOR activator restores S6 expression. Further experiments are also needed to quantify the effects of nutrient starvation on mTOR-mediated cell survival and proliferation, such as by measuring cell death and cell proliferation with respective colourimetric assays.

4.3 Optimisation of glutamate uptake assay to study astrocyte function

To optimise glutamate uptake assay, the specific effect of neuronal co-culturing on glutamate uptake of astrocytes was assessed between co-culture and mono-culture systems to evaluate whether astrocytic glutamate uptake function is promoted by neuronal presence, based on the proposed theory of the functional co-dependency between neurons and astrocytes.

Overall the glutamate uptake assay produced robust results. Astrocytes appeared to be more efficient in taking up extracellular glutamate in neuron-astrocyte co-culture condition than in mono-cultures, evidenced by the significantly smaller co-culture AUCs than mono-culture AUCs, calculated from the drop in extracellular concentrations over time.

AUC is negatively correlated to the rate of glutamate uptake, as it can generally be understood by: $AUC \approx \Delta \text{glutamate concentration} \times \text{time}$. For a constant overall change in glutamate concentration, the smaller the AUC, the shorter the time required for the glutamate uptake, implicating a higher uptake rate.

This signifies that neurons facilitate the glutamate uptake function in co-cultures with astrocytes, which agrees with the tripartite synapse theory, where astrocytes are in physical proximity with the synapse to aid in synaptic clearance of glutamate (Ji et al., 2019). This also aligns with previous findings regarding not only the interaction between neurons and astrocytes, but also the increasing evidence for neurons supporting astrocytic function and vice versa, implicating a co-dependent nature (Brown, 2000).

An alternative explanation to the higher rate of glutamate uptake observed in co-cultures could be neuronal uptake on top of astrocytic uptake, which could be supported by the ability of neurons to switch to glutamate oxidation as an alternative to glucose metabolism conventionally thought to be the major energy fuel, in avoidance of excitotoxic stress (Fendt & Verstreken, 2017). This is however unlikely as EAATs that mediate glutamate transport are predominantly expressed in glia cells of the CNS, astrocytes in particular (Danbolt, 2001; Parkin et al., 2018). This could be further evaluated by measuring the EAAT expression levels in the neuronal population using qRT-PCR.

The change of extracellular glutamate concentrations recorded over time were to be normalised to the respective protein concentrations, in order to obtain the glutamate uptake normalised to the number of functional astrocytes. To provide an estimate of the number of astrocytes, the plan was to isolate astrocytes from the co-culture and using a BCA assay to measure the protein level.

However, because fluorescence-activated cell sorting (FACS) could not be carried out on the co-cultures, a BCA assay to determine protein concentrations of astrocytes could not be conducted either. Therefore, only the absolute change of non-normalised glutamate concentrations over time was shown in Figure 10.

Also, there were variable numbers of data points per time point, because data were gathered from multiple rounds of glutamate uptake assay that inadvertently had different series of timepoints.

Potential improvements

To further enhance the functionality of rat astrocytes, dibutyryl cyclic AMP (dBcAMP) could be added to promote glutamate transporter expression in astrocytes, which had been shown to result in an increase in glutamate uptake V_{max} (Swanson et al., 1997). This could also be included in the further optimisation of the assay. Different initial glutamate concentrations could be assessed, so that the protein concentrations-normalised glutamate uptake could be plotted against different initial glutamate concentrations to obtain glutamate uptake V_{max} and glutamate K_m values for effective comparison with other studies in the literature.

Relying on the sodium-potassium electrochemical gradient established by the sodium-potassium ATPase, sodium-dependent EAATs co-transport one glutamate molecule with three sodium ions (Danbolt, 2001). Therefore, sodium uptake could also be measured to check for co-transport with glutamate to confirm that the glutamate uptake is mediated by EAAT1 or EAAT2.

4.4 Limitations

There were a range of limitations caused by the COVID-19 pandemic and the lack of practical experience at the start of the project. COVID-19 has been a huge disruption to the execution of many details of the original project plan; it has also led to an overall loss of the second part of original project plan, which involves two sub-parts:

- 1) Differentiating TSC2 WT and KO hESCs into astrocytes with the more suited astrocyte differentiation protocol, after trialling the protocols with iPSCs
- 2) Examining the effects of nutrient supply on the mTOR pathway and glial function in these differentiated astrocytes *in vitro*, after optimising the assays using rat astrocytes

All other experimental limitations encountered in this project have been summarised in Table 12, including limitations that were already discussed and not discussed in the above text.

Technique	(N/C) Reason	Limitation
Cell culture	N	- Astrocyte differentiation: difficulty in maintaining consistency in time points used for both differentiation rounds as confluency was reached at different rates for each passaging
	C	- Astrocyte differentiation: results lack robustness due to inability to carry out two differentiation rounds for each protocol
	N	- Neurone-astrocyte co-culture: higher seeding density for astrocytes would have been more ideal (> 15000 cells/cm ²) because they do not proliferate in serum-free medium, and serum is absent in BrainPhys medium
IF	C	- inconsistent, low number of captured IF images per time point per protocol due to limited access to the microscopy facility
	N	- lack of preparation of unstained negative controls for all samples resulting in high auto-fluorescence in IF images with few to zero stained cells
qRT-PCR	N	- lack of preparation of positive controls for all markers
	N	- inconsistency of preparing samples in triplicates (n=3) for all samples
Western blotting	N	- low amounts of purified protein (particularly for serum-starved cultures)
Glutamate uptake assay	N	- inconsistency in time point series in each experimental run
	C	- unable to carry out FACS on co-cultures to sort out astrocytes from neurons
	C	- unable to conduct BCA assay to determine protein concentrations for normalising glutamate uptake
Overall	N/C	- low number of experimental runs for most conditions for assays (n≈3)

Table 12: Summary of all non-COVID-19-related and COVID-19-related experimental limitations non-COVID-19-related (N), COVID-19-related (C).

5 Conclusion

Both astrocyte differentiation using STEMCELL Technologies Kits and by the serum-driven method did not show promising results in generation of mature human iPSC-derived astrocytes. In rat astrocytes, serum starvation impaired S6 expression; however, starvation of both glucose and serum did not lead to a significant decrease in the mTORC1 activation for both 24- and 72-hour medium incubation. For the optimisation of the glutamate uptake assay, neuron-astrocyte co-cultures showed a higher rate of glutamate uptake than mono-cultures.

In addition to the limitations encountered in this project, the variability in certain data sets and the generally low number of samples also imply that the results should be interpreted with caution.

Overall, much of the scope of astrocytic research is not well established and requires standardisation and refining. More endeavours in researching the mTOR pathway and astrocytes will hopefully increase the understanding of their roles and uncover potential therapeutic targets for a range of diseases such as TSC.

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